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Rockchip RV1103G_RV1106G

Hardware Design Guide

(Fuzhou Hardware Development Center)

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Preface

Overview

This document presents the key points of hardware design and notices for RV1103G/RV1106G processors, aiming to help customers shorten developing period, improving product design stability and reducing fault rate. Please refer to the requirements of this guide for hardware design, and use the relevant core templates released by Rockchip. If you need to modify for special reasons, please strictly follow the design rule of high-speed-digital-circuit and Rockchip Schematic&PCB checklist requirements.

Chipset model

This document is suitable for the following chipset model: RV1106G, RV1103G

Intended Audience

This this guide is mainly intended for:

- Hardware development engineers
- Layout engineers
- Technical support engineers
- Test engineers

Revision History

This revision history recorded description of each version, and any updates of previous versions are included in the latest one.

Version No.	Author	Revision Date	Revision Description	Remark
V1.0	Linus.Lin	2022.06.10	English version initial released (CN V1.1)	
V1.1	Linus.Lin	2022.12.21	Correct the “release the button to make ADCIN1 return to high level (1.8V)” to “release the button to make ADCIN0 return to high level (1.8V)” at 2.3.8	

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1 System Introduction

1.1 Overview

RV1106 is a highly integrated vision processor SoC, especially for AI related application.

It is based on single-core ARM Cortex-A7 32-bit core which integrates NEON and FPU. There is a 32KB I-cache and 32KB D-cache and 128KB unified L2 cache. The build-in NPU supports INT4/INT8/INT16 hybrid operation and computing power is up to 0.5TOPs. In addition, with its strong compatibility, network models based on a series of frameworks such as TensorFlow/MXNet/PyTorch/Caffe can be easily converted. RV1106 introduces a new generation totally hardware-based maximum 5-Megapixel ISP (image signal processor). It implements a lot of algorithm accelerators, such as HDR, 3A, LSC, 3DNR, 2DNR, sharpening, dehaze, gamma correction and so on. Cooperating with two MIPI CSI (or LVDS) and one DVP (BT.601/BT.656/BT.1120) interface, users can build a system that receives video data from multiple camera sensors simultaneous.

1.2 Chip Differences

Table 1-1 RV110x Chip Differences

	RV1103G1	RV1106G2
Sensor	MAX 4M	MAX 5M
DDR	64MB	128MB
Package	QFN88	QFN128
Sensor In	MIPI: 1*4lane (2*2lan) maximum 2-way, 2lane*MIPI+2lane*MIPI	MIPI: 1*4lane or 2*2lane VICAP: 16bit maximum 3-way, 2lane*MIPI+2lane*MIPI+10bit DVP
FSPI	FSPI	FSPI or 8bit eMMC
EMMC		
SDMMC	1	1
SDIO	1	1
BT	NA	support
LCDC/BT1120	NA	18bit
RTC	NA	build-in
LPMCU	NA	build-in

1.3 Block Diagram

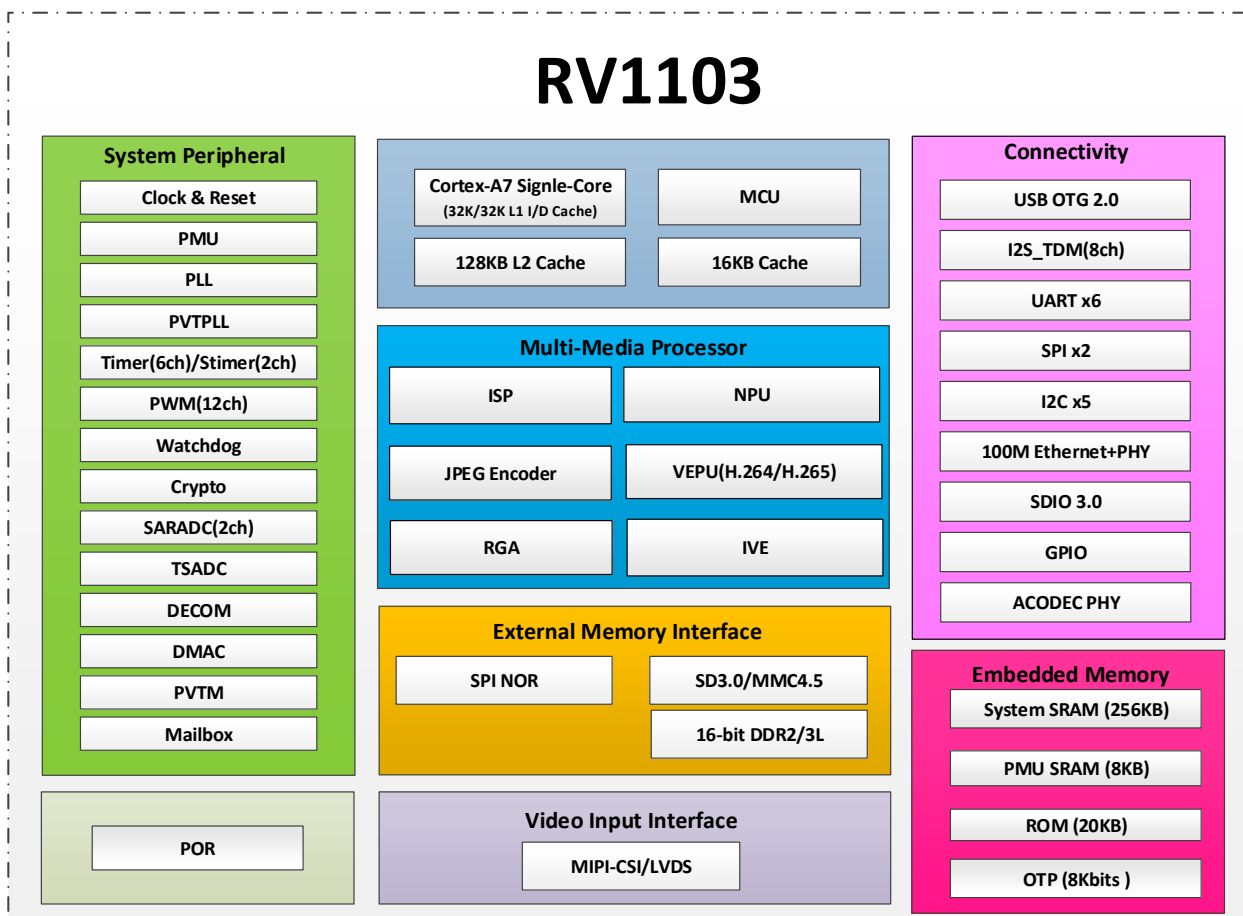


Figure 1-1 RV1103G block diagram

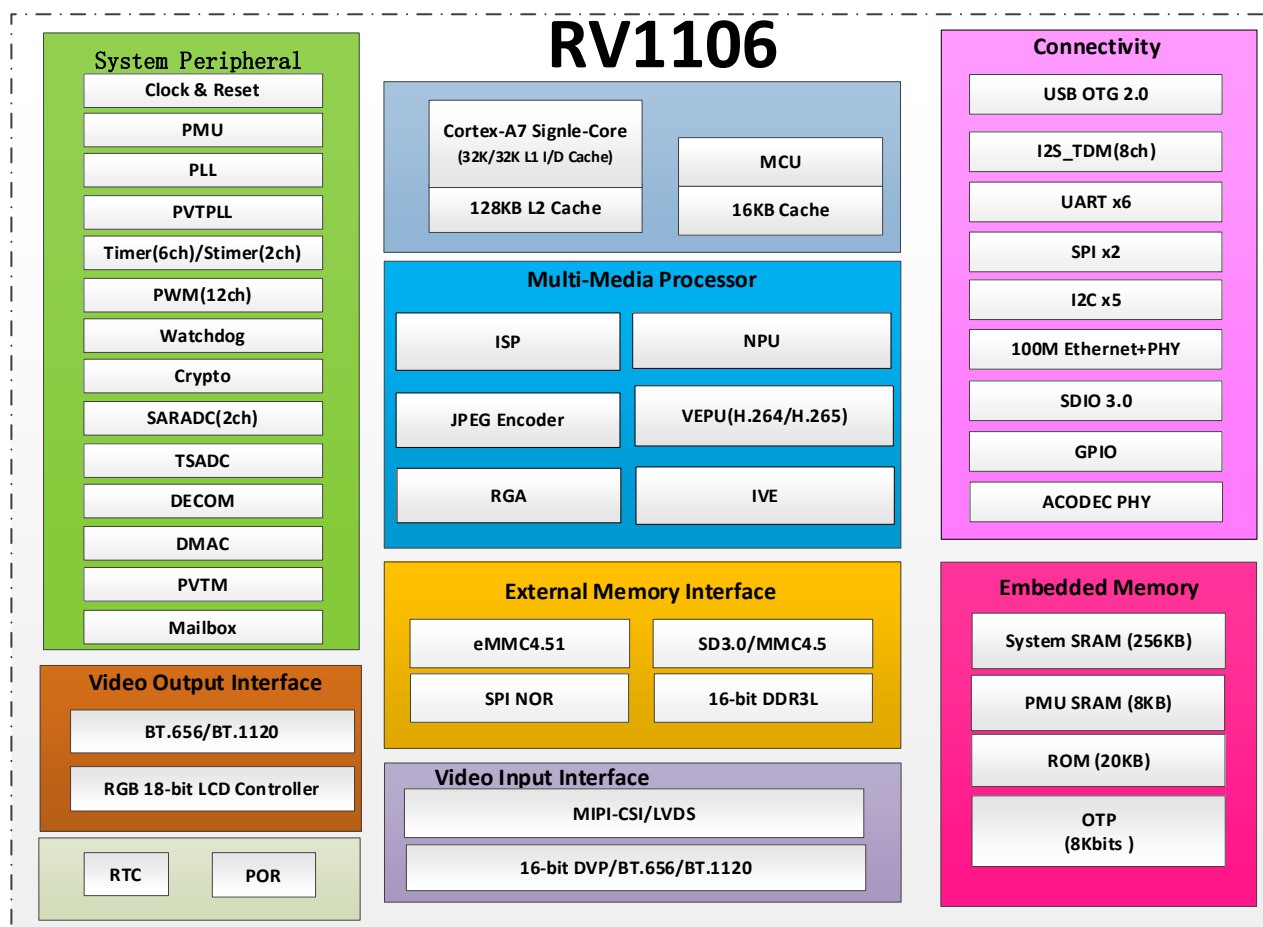


Figure 1-2 RV1106G block diagram

1.4 Application Block Diagram

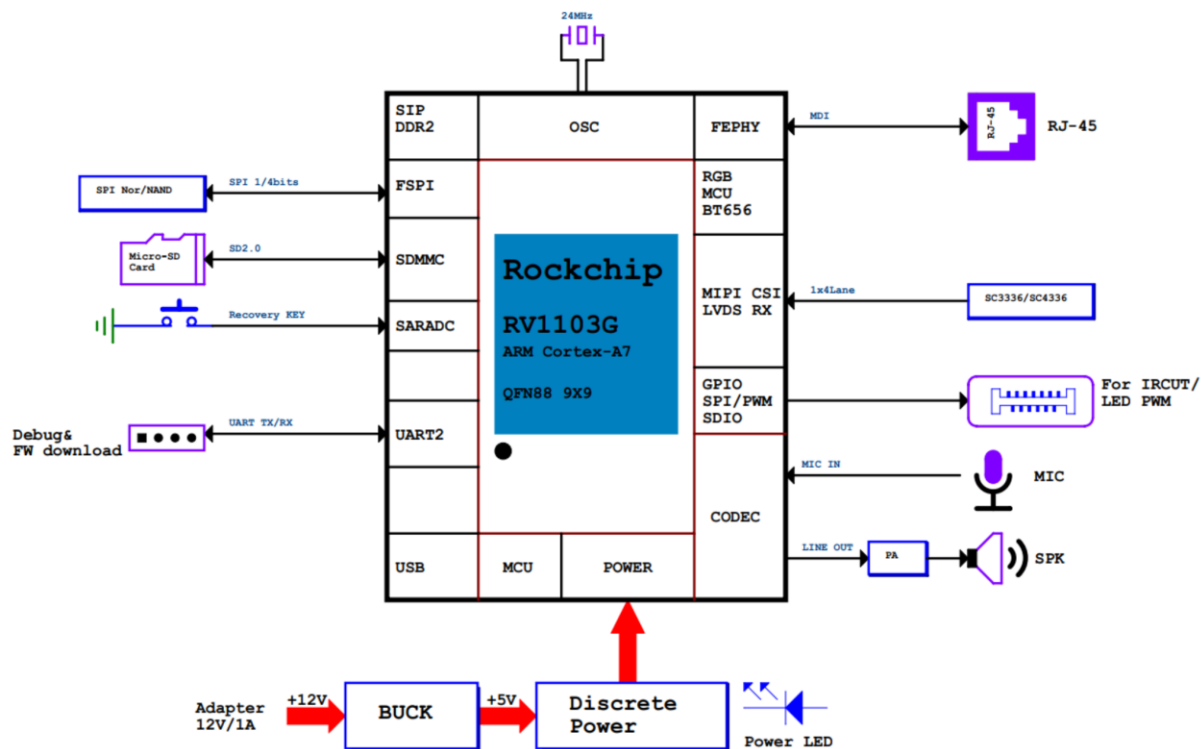


Figure 1-3 RV1103G application block diagram

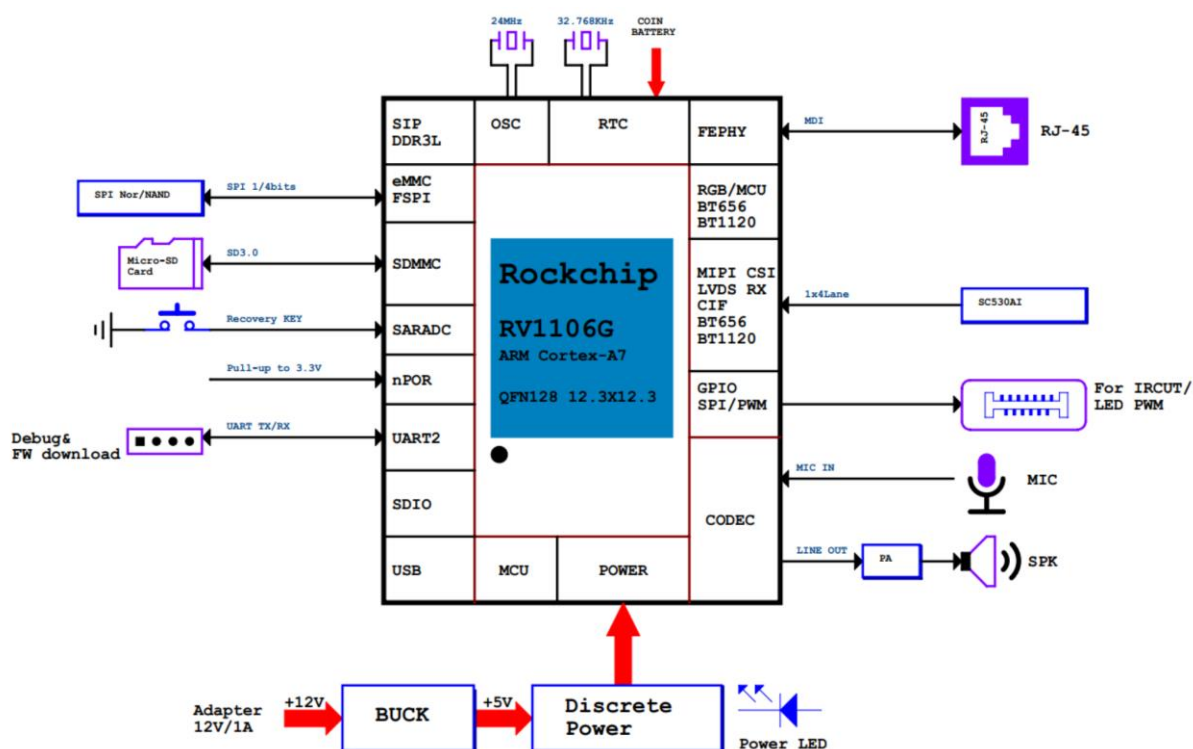


Figure 1-4 RV1106G application block diagram

2 Schematic Design Recommendation

2.1 Minimum System Design

2.1.1 Clock Circuit

The oscillator circuit inside RV1103G/RV1106G and the external 24MHz crystal form the system clock, as shown in Figure 2-1.

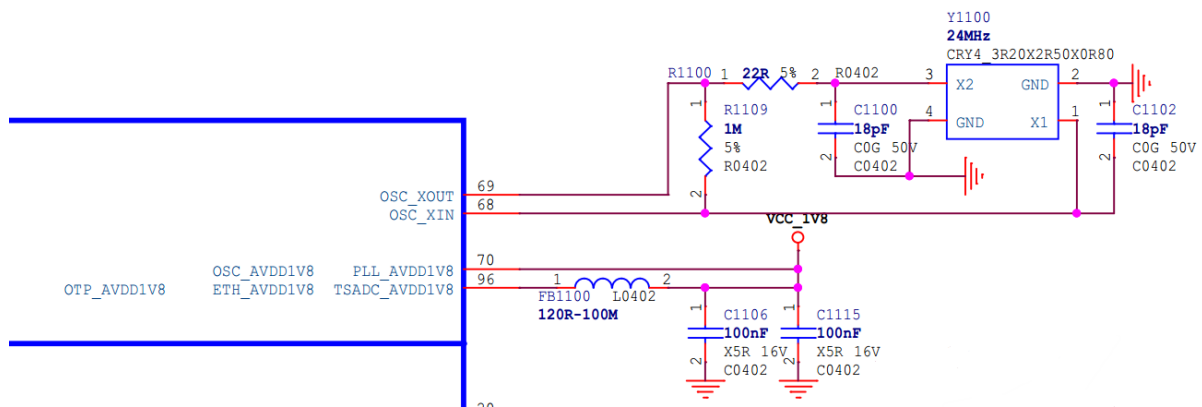


Figure 2-1 RV1103G/RV1106G Crystal Circuit



NOTE

The selected capacitor needs to match the load capacitance of the crystal oscillator, which the material is recommended to be NPO. It is recommended to use a 4Pin chip crystal oscillator, of which 2 GND pins are fully connected to the ground to enhance the anti-ESD interference capability of the system clock. Use a crystal with a frequency offset less than 20ppm.

The system clock can also be directly generated by an external active crystal circuit and input through the OSC_XIN pin. The clock parameters are shown in Table 2-1 below:

Table 2-1 RV1103G/RV1106G 24MH clock requirements

Parameters	Spec			Description
	Min.	Max.	Unit	
Frequency	24.000000		MHz	
Frequency tolerance	+/-20		ppm	Frequency tolerance
Clock amplitude	1.8V±10%		V	

Compared with RV1103G, RV1106G has one more RTC. The circuit is shown in Figure 2-2 below. Pay special attention to the RC parameters of the AVDD33_RTC power input, which **must be 100ohm and 1uF**.

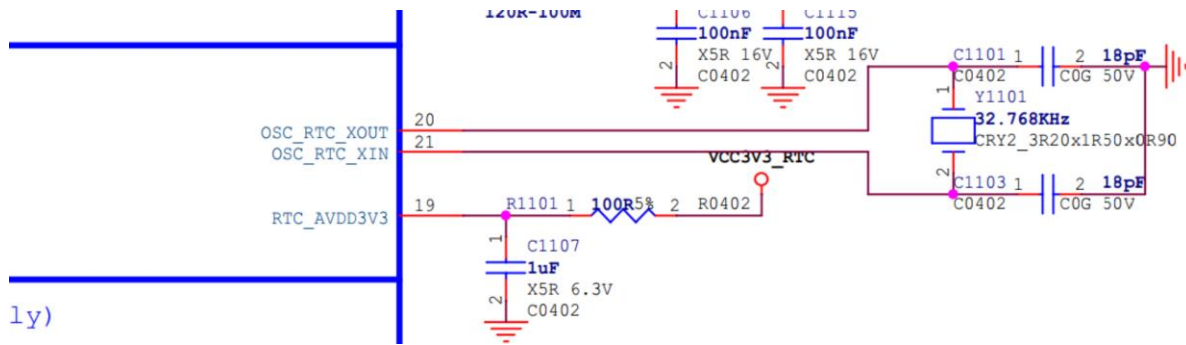


Figure 2-2 RV1106G RTC crystal circuit

**NOTE**

RV1103G does not support RTC.

The selected capacitor need to match the load capacitance of the crystal oscillator, which the material is recommended to be NPO.

2.1.2 NPOR Circuit

The RV1103G/RV1106G chip integrates a POR (Power on Reset) circuit. As shown in Figure 2-3, the chip pin66 is pulled up to the PMU_VCC3V3 power supply through the resistor R1110, which is used as the input reference of the voltage comparator of the internal NPOR. The capacitor C1114 is used to eliminate jitter. Although the RV1103G has no NPOR pin, it is pulled up to the PMU_VCC3V3 power supply inside the chip as the input reference of the voltage comparator. When laying out, place the resistor and capacitor close to the RV1106G.

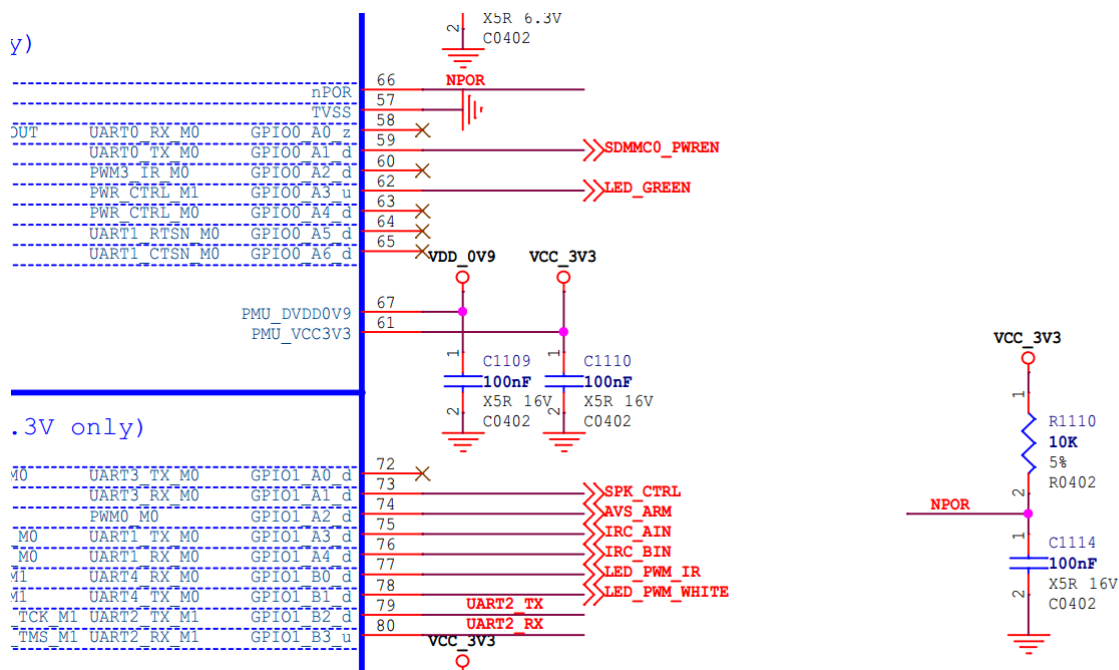


Figure 2-3 RV1106G NPOR input

2.1.3 System Boot Sequence

The system boot sequence of the RV1103G/RV1106G chip has the following priorities from high to low:

- SFC/SPI FLASH
- eMMC FLASH
- SDMMC CARD

2.1.4 System Initializes Configuration Signal

RV1103G/RV1106G multiplexes JTAG function with SDMMC function to reduce IO extraction and switch output mode through pins. This function is mainly convenient for users to debug the system through the SD card interface without disassembling the machine after the whole machine is installed. Pin configuration as the follow table:

Table 2-2 RV1103G/RV1106G System Initialization Configuration Signal Description

Signal name	Internal up and down	Description
SDMMC0_DET	Pull up	JATG pin multiplexing selection control signal: 0: Recognized as SD card insertion, SDMMC/JATG/UART pins are multiplexed as SDMMC output; 1: Recognized as SD card not inserted, SDMMC/JATG/UART pins are multiplexed as JTAG/UART output (Default).

2.1.5 JTAG Debug Circuit

The RV1103G/RV1106G's JTAG interface conforms to IEEE1149.1 standard. PC connect the DSTREAM emulator through the SWD mode (two-wire mode) to debug the ARM Core and MUC Core inside the chip.

As mention above, it is necessary to ensure the SDMMCO_DET pin is high-level before connecting the emulator, otherwise the JTAG debugging mode cannot be entered. Interface description as follow:

Table 2-3 RV1103G/RV1106G JTAG Debug interface signal

Signal name	Description
JTAG_ARM_TCK	ARM Cortex-A7 JTAG clock input
JTAG_ARM_TMS	ARM Cortex-A7 JTAG mode select input
JTAG_LPMCU_TCK	Low Power MCU JTAG clock input
JTAG_LPMCU_TMS	Low Power MCU JTAG mode select input
JTAG_HPMCU_TCK	High Performance MCU JTAG clock input
JTAG_HPMCU_TMS	High Performance MCU JTAG mode select input

**NOTE**

RV1103G has no LPMUC.

2.1.6 FSPI Circuit

FSPI (Flexible-SPI) is a flexible serial peripheral interface host controller that interfaces with external devices. It has the following characteristics:

- Support serial NOR and NAND FLASH
- Support SDR mode. Support single/dual/four-wire mode

The recommended design for pull-up and down and matching design of the SPI interface as show in the Tab.2-4:

Table 2-4 RV1103G/RV1106G SPI interface desgin

Signal	Internal pull up/down	Connection mode	Description(chipset)
FSPI_D[3:0]	pull up	direct connection	FSPI data send/receive
FSPI_CS0	pull up	direct connection	SPI chip select signal
FSPI_CLK	pull down	connect 22ohm resistor in series	SPI clock send

2.1.7 eMMC Circuit

2.1.7.1 eMMC Controller Introduction

The RV1106G eMMC interface supports eMMC 4.51 and is compatible with 4.41 and 5.0 protocol devices. The controller has the following features:

- Support 1-bit, 4-bit and 8-bit data bus width;
- Up to HS200 mode is supported, but CMD Queue is not supported.

2.1.7.2 eMMC Topology and Connect Method

The recommended design for pull-up and down and matching design of the eMMC interface as show in the Tab.2-5:

Table 2-5 RV1106G eMMC interface design

Signal	Internal pull up/down	Connection mode	Description(chipset)
eMMC_DQ[7:0]	pull up	direct connection	eMMC data send/receive
eMMC_CMD	pull up	direct connection	eMMC command sent/receive
eMMC_CLK	pull down	connect 22ohm resistor in series	eMMC clock sent

2.1.7.3 eMMC Power Up Sequence Requirement

The RV1106G chip eMMC controller belongs to the VCCIO3 power domain:

- GPIO3: I/O power supply of eMMC controller.

The eMMC have two sets of power supplies, refer to JEDEC standard for power-on sequence:

- There is no requirement for VCC and VCCQ in the power-on sequence;
- VCC and VCCQ must be powered on before the CMD command of RV1106G is issued, and maintain a stable working voltage;
- After the chip enters sleep mode, RV1106G can turn off the VCC power supply to reduce power consumption;
- Before the chip wakes up from sleep mode, the VCC power supply must be powered on and maintain a stable working voltage.

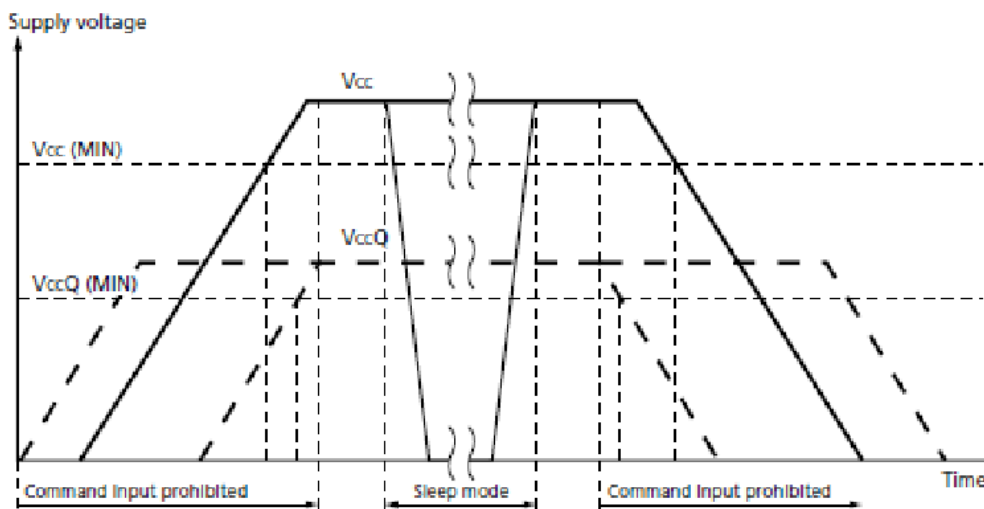


Figure 2-4 eMMC Power Up and Power Down Sequence

2.1.7.4 eMMC Support List

For RV11026G eMMC support list, please refer to the document "RKeMMCSupportList", which can be downloaded from Rockchip redmine platform:

https://redmine.rockchip.com.cn/projects/fae/documents?tdsourcetag=s_pctim_aiomsg



NOTE

RV1103G does not support eMMC.

2.1.8 GPIO Circuit

In RV1103G/RV11026G, there are two types of GPIO: 1.8V and 3.3V, and supports self-adaption.

2.1.8.1 GPIO Drive Capability

In the RV1103G/RV1106G chip, GPIO provides 6 adjustable drive strengths, which are 2mA/4mA/6mA/8mA/10mA/12mA. The initial default drive strength is different according to the type of GPIO. Please refer to the chip TRM for configuration modification.

2.1.8.2 GPIO Power

The power pins of GPIO power domain are described as follows:

- When the IO power pin name is PMU_VCC3V3, it means PMU power domain and only supports 3.3V. The GPIO1_VCC3V3 is similar.
- When the IO power name is VCCIO3_VCC, it means VCCIO3 power domain and supports 3.3V or 1.8V. VCCIO5_VCC is similar.
- When the IO power pin name is MIPI_VCCIO7_AVDD1V8, it means VCCIO7 power domain and only supports 1.8V.
- It is recommended to place a 100nF decoupling capacitor on each power supply pin of the power domain and place it close to the power supply pin. For detailed design, please refer to the RV1103G/RV1106G chip reference schematic diagram.
- The power supply ripple is required to be $\pm 5\%$.

Table 2-6 RV1103G/RV1106G GPIO power pins description

Power domain	GPIO Type	Pin name	Description
PMU	3.3V only	PMU_DVDD0V9	0.9V logic power for this GPIO domain.
		PMU_VCC3V3	3.3V IO supply for this GPIO domain.
VCCIO1	3.3V only	VCCIO1_VCC3V3	3.3V IO supply for this GPIO domain.
VCCIO2	1.8V only	SARADC_USB_AVDD1V8	1.8V IO supply for this GPIO domain.
VCCIO3	1.8V/3.3V	VCCIO3_VCC	1.8V or 3.3V IO supply for this GPIO domain.
VCCIO4	3.3V only (RV1103G)	VCCIO4_VCC3V3	3.3V IO supply for this GPIO domain.
	1.8V/3.3V (RV1106G)	VCCIO4_VCC	1.8V or 3.3V IO supply for this GPIO domain.
VCCIO5	1.8V/3.3V	VCCIO5_VCC	1.8V or 3.3V IO supply for this GPIO domain.
VCCIO6	1.8V/3.3V	VCCIO6_VCC	1.8V or 3.3V IO supply for this GPIO domain.
VCCIO7	1.8V only	MIPI_VCCIO7_AVDD1V8	1.8V IO supply for this GPIO domain.

**NOTE**

RV1103G has no GPIO5 power domain.

RV1103G VCCIO4 only supports 3.3V mode.

2.2 Power Supply Design

2.2.1 Power Up Sequence

The power-on sequence of the RV1103G/RV1106G chip has the following requirements:

- It follows the principle that the same module is powered up first with low voltage and then powered up with high voltage; the different module with the same voltage is powered up at the same time.
- Different timing intervals must be greater than 1ms.
- The soft-start time of the power device must be greater than 100us. Can't use a device with a soft-start time of less than 100us.

The recommended power-up sequence is as follows:

VDD_0V9&VDD_ARM-→VCC_1V8&VCC_DDR-→VCC_3V3

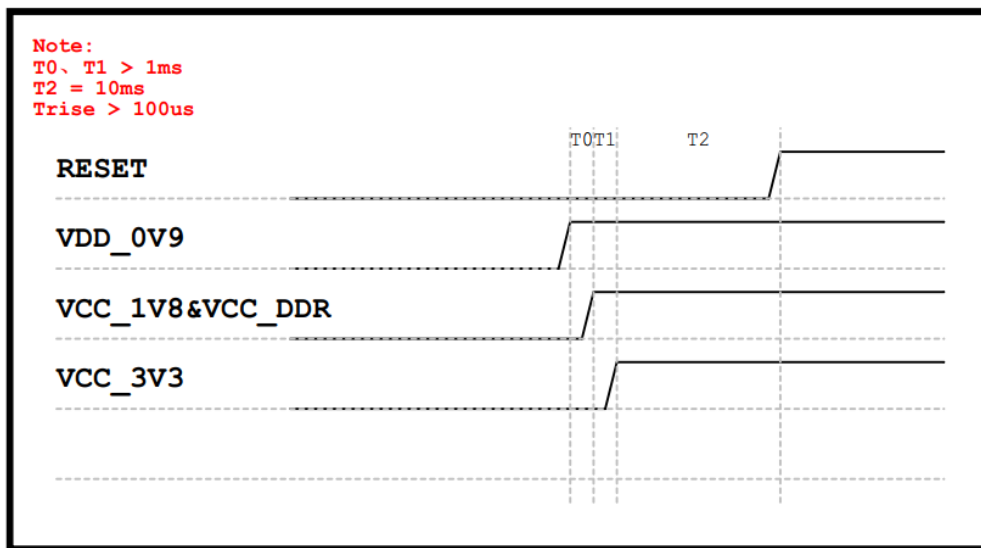


Figure 2-5 RV1103G power up sequence

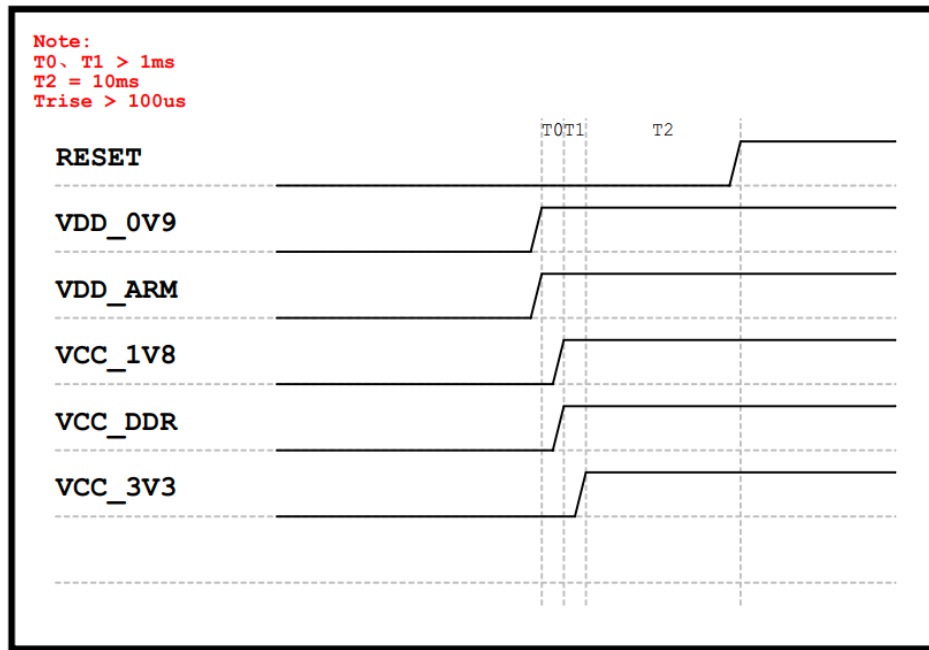


Figure 2-6 RV1106G power up sequence

2.2.2 Power Off Sequence

There is no strict requirement for the power-off sequence of the RV1103G/RV1106G, and all power sources can be powered off at the same time.

2.2.3 Power Supply Design Suggestion

2.2.3.1 CPU Power Supply

RV1103G/RV1106G uses CPU independent power domain for power supply. As shown in the Fig.2-8, VDD_ARM supplies power for ARM Cortex-A7 core. It supports DVFS dynamic frequency modulation and voltage regulation function, and uses DC-DC power supply for independent power supply with a peak current of 400mA. The capacitors in the RV1103G/RV1106G chip reference design schematic cannot be omitted. Place the capacitor close to the chip pins during layout, and the power supply ripple is required to be $\pm 5\%$. The capacitor is shown in Figure 2-8.



Figure 2-7 RV1103G/RV1106G VDD_CPU power

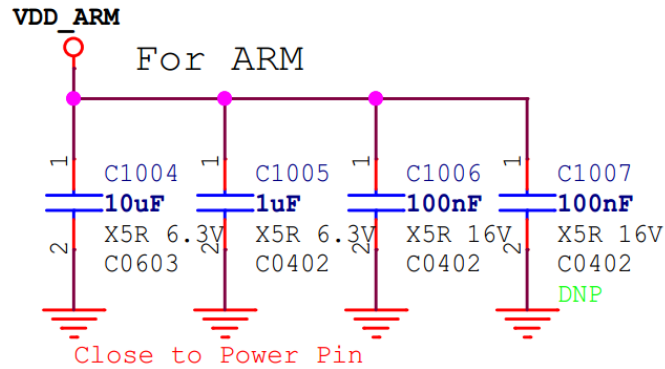


Figure 2-8 The decouple of RV1103G/RV1106G VDD_CPU power

2.2.3.2 LOGIC&NPU Power Supply

The RV1103G/RV1106G GPU & LOGIC power supply is powered by DC-DC separately, as shown in the Fig.2-9 VDD_0V9, and the peak current can reach 1.6A, so please do not delete the capacitors in the schematic diagram. When Layout, please place 2 10uF capacitors on the near power supply side and the far power supply side of the RV1103G/RV1106G chip respectively. Each pin should have at least 1 100nF decoupling capacitor placed close to the chip pins. The power supply ripple is required to be within $\pm 5\%$, the capacitance is shown in Fig.2-10.

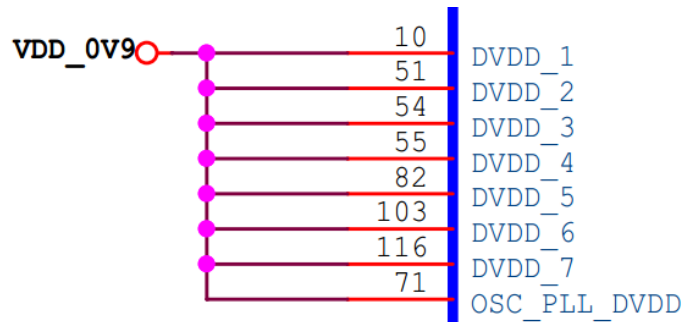


Figure 2-9 RV103G/RV1106G LOGIC&NPU Power

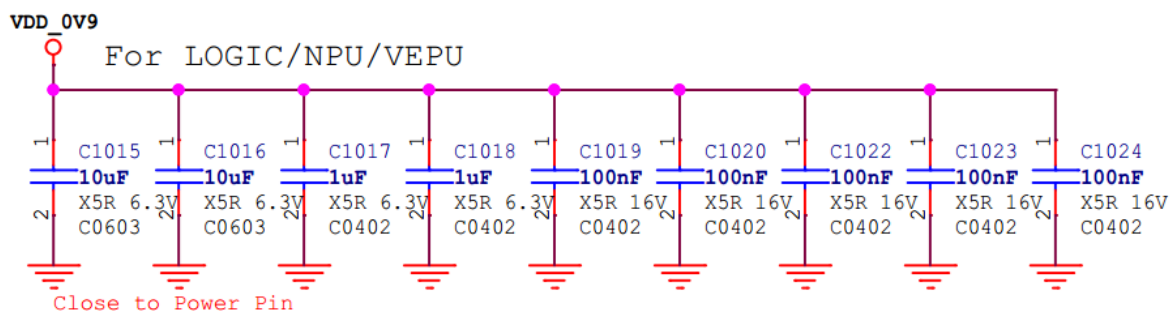


Figure 2-10 The decouple of RV1103G/RV1106G LOGIC&NPU power

2.2.3.3 DDR Power Supply

The RV1103G/RV1106G DDR power is powered by DC-DC separately, and the peak current is 400mA. Do not omit the capacitors in the RV1103G/RV1106G chip reference design schematic. The power supply ripple is required to be $\pm 5\%$. Place capacitors close to the chip pins during layout.

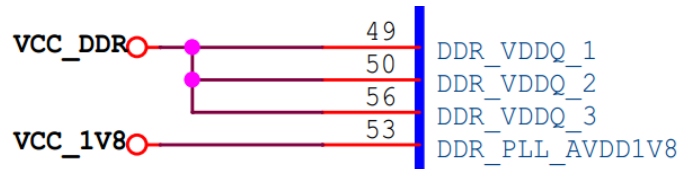


Figure 2-11 RV1103G/RV1106G DDR controller power

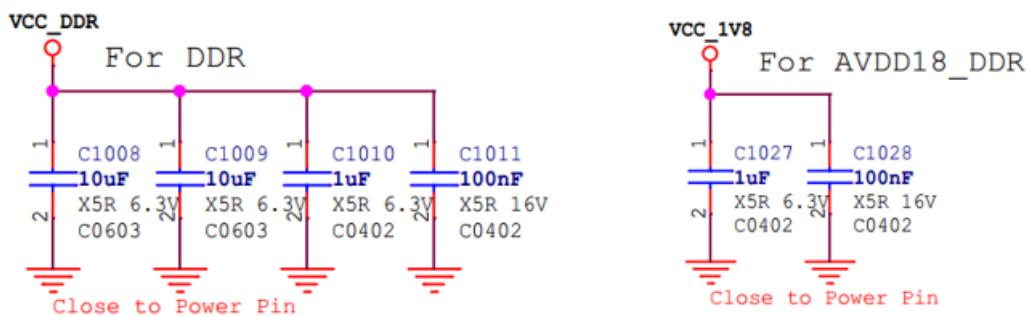


Figure 2-12 The decouple of RV1103G/RV1106G DDR power

2.2.3.4 GPIO Power Supply

Please refer to Section 2.1.8 for GPIO power supply. It is recommended to place 1 100nF decoupling capacitor per pin and place it close to the power supply pins. For detailed design, please refer to the RV1103G/RV1106G reference design schematic.

2.3 Functional Interface Circuit Design Guide

2.3.1 SDMMC Interface

RV1103G/RV1106G provides an SDMMC interface controller that supports SD V3.0 (RV1103G only supports SD V2.0) and MMC V4.51 protocol, as shown in Fig.2-13:

- The SDMMC controller is powered by a separate power domain;
- SDMMC and UART2, JTAG and other functions are multiplexed together, and function selection is performed through SDMMC0_DET. For details, please refer to Section 2.1.4;
- VCCIO4_VCC is the IO power supply, which requires external 3.3V power supply (SD 2.0 mode) or 3.3V/1.8V adjustable power supply (SD 3.0 mode);

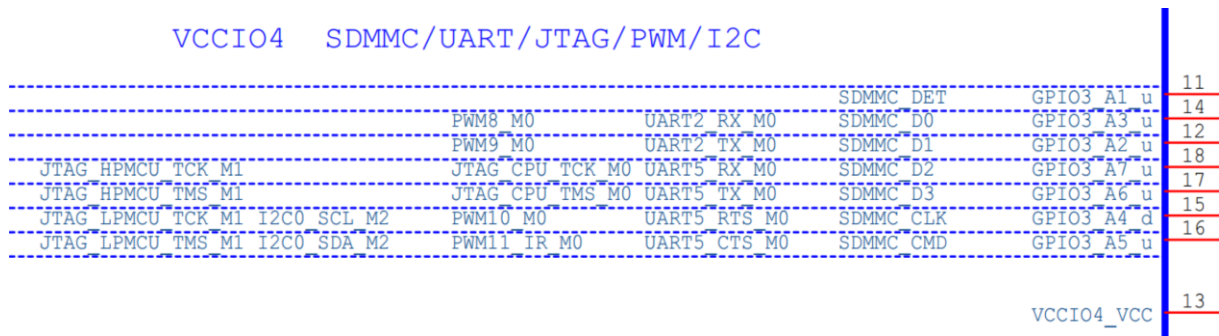


Figure 2-13 RV1103G/RV1106G SDMMC module circuit

The design recommendations for the pull-up and down and matching design of the SDMMC interface as show in Tab.2-7:

Table 2-7 RV1103G/RV1106G SDMMC Interface Design

Signal	Internal pull-up and pull-down	Connection mode (SDR104 high speed mode)	Description (chipset)
SDMMC_DQ [3:0]	Pull up	Direct connect	SD data sent/receive
SDMMC_CMD	Pull up	Direct connect	SD command sent/receive
SDMMC_CLK	Pull down	Connect a 22ohm resistor in series	SD clock sent

2.3.2 SDIO/UART Interface

RV1103G/RV1106G supports WIFI/BT modules with SDIO 3.0 interface, as shown in Fig.2-14. When using WIFI/BT modules with SDIO and UART interfaces, it should be noted that the IO power supply VCCIO5_VCC of the controller must be consistent with the IO level of the module.

Considering the flexibility of the application, SDIO has two multiplexing M0 and M1, which are located in the VCCIO5 and VCCIO6 power domains respectively, and only one of them can be used at the same time. As shown in Fig.2-14 and Fig.2-15.

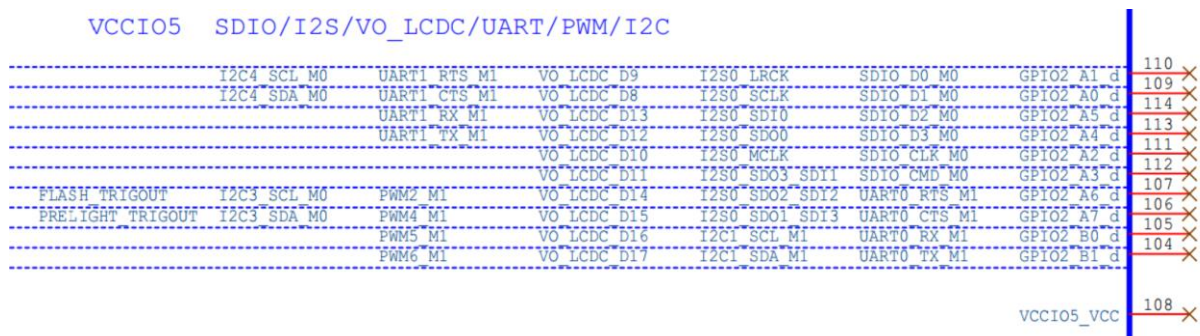


Figure 2-14 RV1106G SDIO multiplexing interface 1

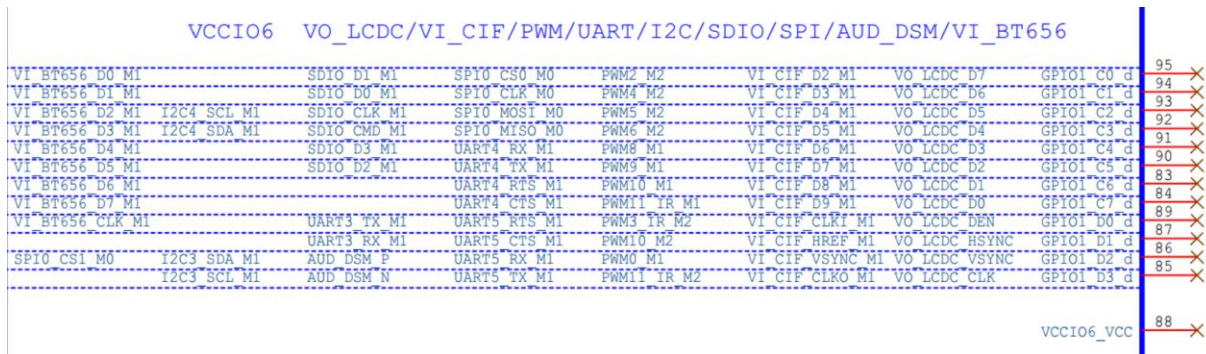


Figure 2-15 RV1103G/RV1106G SDIO multiplexing interface 2

2.3.2.1 SDIO

The design recommendations for the pull-up and down and matching design of the SDIO interface as show in Tab.2-8:

Table 2-8 RV1103G/RV1106G SDIO Interface Design

Signal	Internal pull-up and pull-down	Connection mode	Description (chipset)
SDIO_DQ [0:3]	Pull down	Direct connect	SD data sent/receive
SDIO_CMD	Pull down	Direct connect	SD command sent/receive
SDIO_CLK	Pull down	Connect a 22ohm resistor in series	SD clock sent

2.3.2.2 UART

The design recommendations for the pull-up and down and matching design of the UART interface as show in Tab.2-9:

Table 2-9 RV1103G/RV1106G UART Interface Design

Signal	Internal pull-up and pull-down	Connection mode	Description (chipset)
UART0_RX	Pull up	Direct connect	UART0 data input
UART0_TX	Pull up	Direct connect	UART0 data output
UART0_CTSn	Pull up	Direct connect	UART0 permit sent signal
UART0_RTSn	Pull up	Direct connect	UART0 request sent signal

2.3.3 USB Interface

RV1103G/RV1106G has a group of USB 2.0 interface, which supports HOST and DEVICE mode.

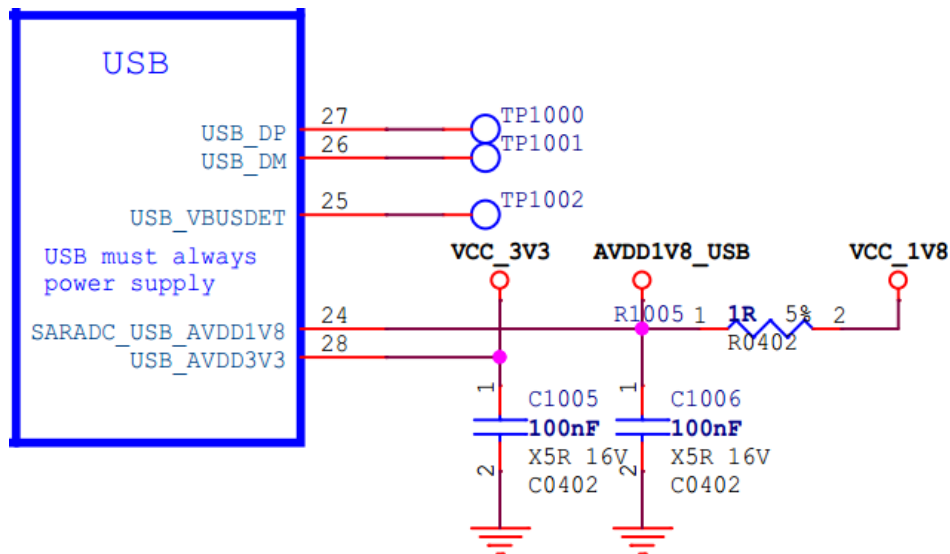


Figure 2-16 RV1103G/RV1106G USB 2.0 module

Something be noted in design:

- If the UART/SD is not used for programming, there need to reserve interface since the USB is used as firmware programming port.

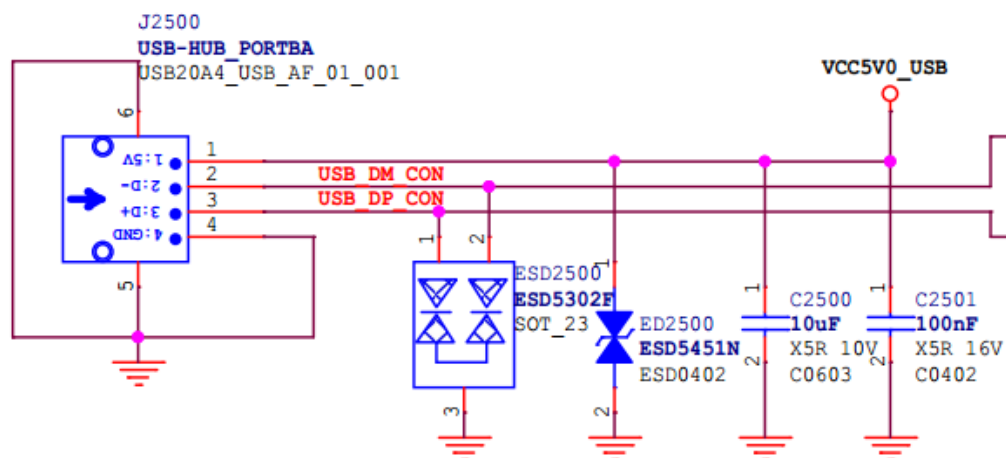


Figure 2-17 RV1103G/RV1106G USB socket

- USB interface is Device mode by default.
- Since the USB_VBUS (USB_DET) is used as the USB device insertion detection, the input detection voltage must be less than 3.3V, and the high level must be detected to be recognized by the computer.

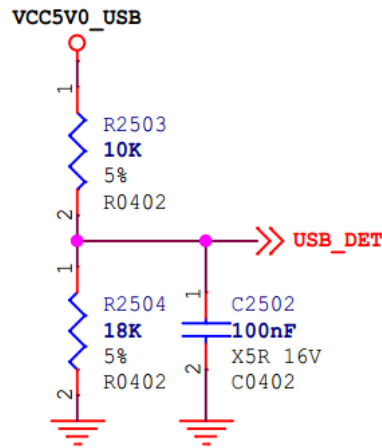


Figure 2-18 RV1106G USB insert detection

**NOTE**

RV1103G don't has the USB insert detection.

- The 1.8V power supply of the controller needs to be connected with a 10hm resistor in series to avoid the damage caused by the surge.

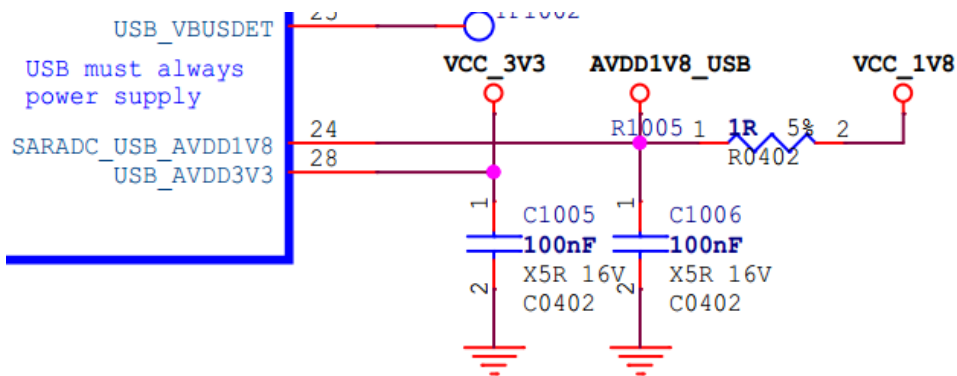


Figure 2-19 RV1103G/RV1106G USB Controller Power Surge Protection

- The decoupling capacitor of controller power need to place close to the pin to enhance USB performance.
- To suppress electromagnetic radiation, consider reserving a common mode choke on the signal line, and choose the resistor or common mode choke according to the actual situation during the debugging process.

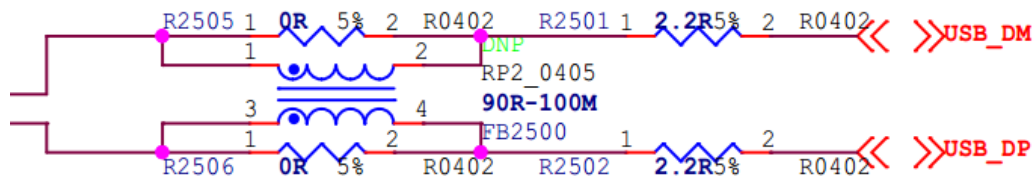


Figure 2-20 RV1103G/RV1106G USB Common mode choke

The design recommendations for the pull-up and down and matching design of the USB 2.0 interface as show in Tab.2-10:

Table 2-10 RV1103G/RV1106G USB interface design

Signal	Connection method	description
USB_DP/DM	Connect a 2.2ohm resistor in series	USB OTG input/output
USB_VBUSDET		USB OTG insert detection

2.3.4 FEPHY Interface

RV1103G/RV1106G integrates a 100M PHY inside, and adds a network transformer and network port connector externally to achieve 100M function.

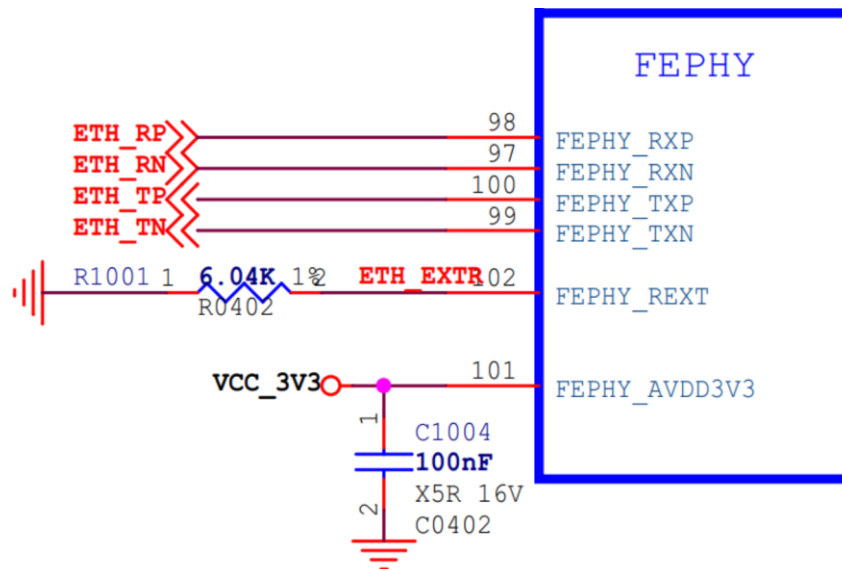


Figure 2-21 RV1103G/RV1106G FEPHY interface

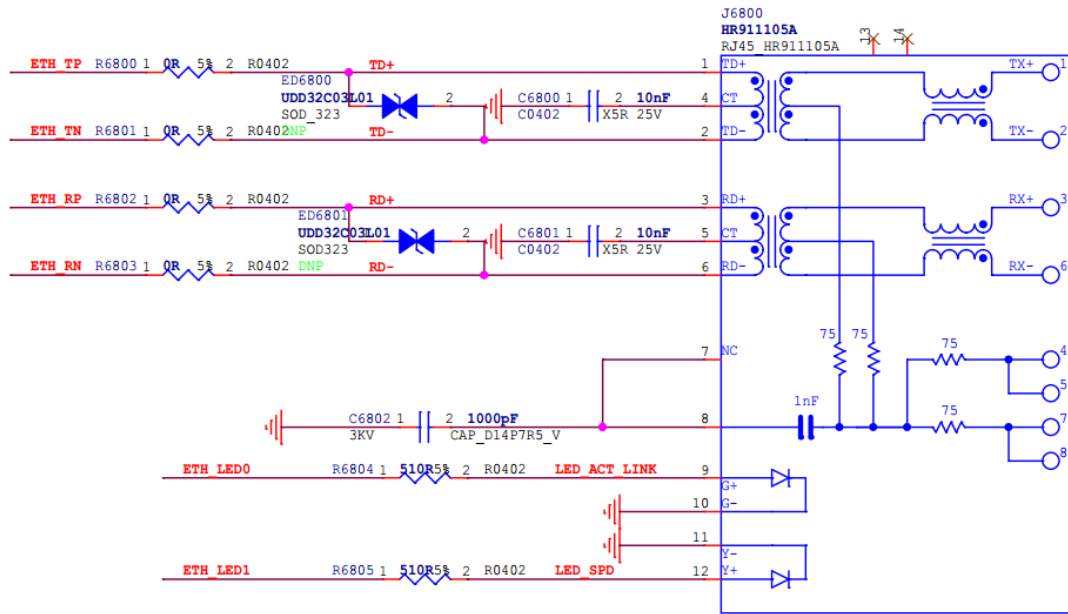


Figure 2-22 RV1103G/RV1106G RJ45 interface

Table 2-11 RV1103G/RV1106G FEPHY

Signal	Internal pull-up and pull-down	Connection mode	Description (chipset)
FEPHY_RXP/_RXN	NA	connect a 0ohm resistor in series	Data transmission differential pair signal
FEPHY_TXP/TXN	NA	connect a 0ohm resistor in series	Data receive differential pair signal
FEPHY_EXTR	NA	Connect 6.04K precision resistor in series to ground	reference resistor

Something be noted in design:

- The 0ohm resistor connected in series on the differential signal cannot be omitted and needs to be reserved.
- The EXTR resistor is placed close to the chip.
- The AVDD18 power supply of FEPHY is susceptible to noise interference from digital signals, so magnetic beads need to be placed nearby. Otherwise, packet loss may occur in poor cable quality or in long cable mode (100m cable).

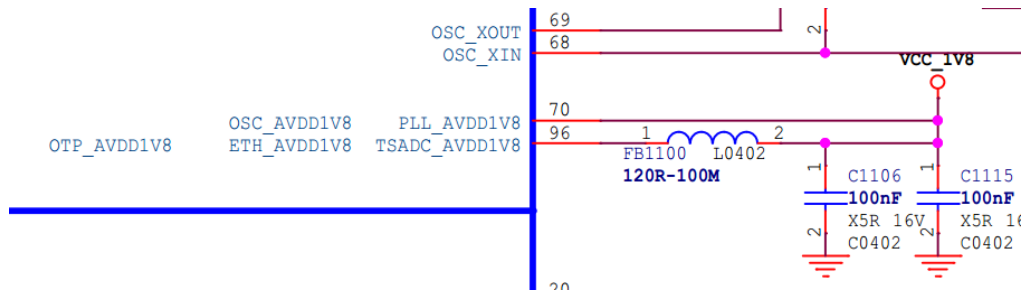


Figure 2-23 RV1103G/RV1106G FEPHY power series magnetic beads

2.3.5 Audio Interface

RV1103G/RV1106G has a built-in ACODEC, and provides a set of standard I2S interfaces for expanding audio equipment, supports master mode, the highest sampling rate is up to 192kHz, and the bit width supports 16bits to 24bits.



NOTE

The built-in ACODEC and the I2S for expansion are a common bus design, so only one of them can be used.

2.3.5.1 ACODEC

The build-in ACODEC of RV1106G supports 1-way lineout output, and 2-ways differential MIC input.

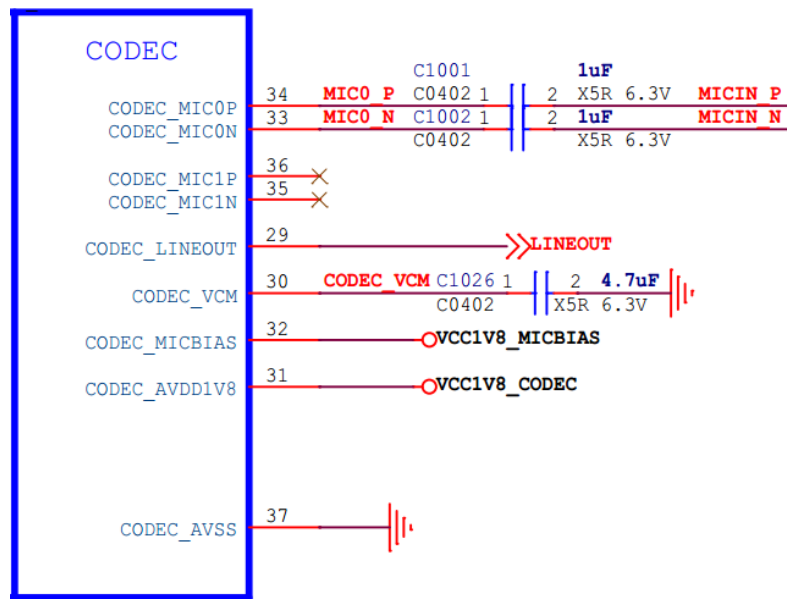


Figure 2-24 RV1106G ACODEC Module

The build-in ACODEC of RV1103G supports 1-way lineout output, and 1-way differential MIC input. The differential MIC can be split into two single-ended input.

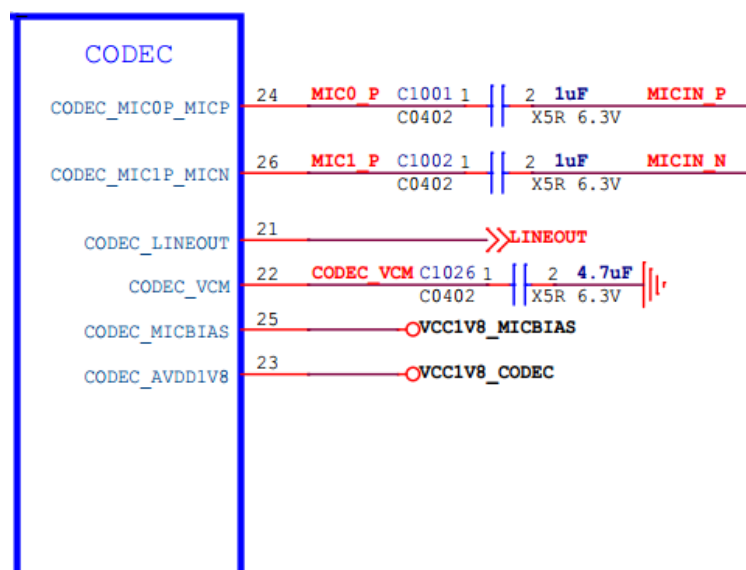


Figure 2-25 RV1103G ACODEC module

The design recommendations for the pull-down and matching design of the ACODEC interface as show in the Tab.2-12.

Table 2-12 RV1103G/RV1106G ACODEC interface design

Signal	Internal pull-up and pull-down	Connection mode	Description (chipset)
CODEC_MIC0P/MIC0N	NA	1uF capacitor in series	MIC0 differential input
CODEC_MIC1P/MIC1N	NA	1uF capacitor in series	MIC1 differential input
CODEC_LINEOUT	NA		LINEOUT output
CODEC_VCM	NA	Connect a 4.7uF capacitor to ground in series	CODEC internal reference
CODEC_MICBIAS	NA		MIC Bias supply output

2.3.5.2 I2S0

As shown in the figure below, the I2S0 interface includes non-independent 8-channel output and 8-channel input, and cannot realize all input and output at the same time. It should be noted that this group of I2S interfaces belong to the GPIO5 power domain, please pay attention to the level matching of peripheral IO.

VO_LCDC_D9	I2S0_LRCK	SDIO_D0_M0	GPIO2_A1_d	110	X
VO_LCDC_D8	I2S0_SCLK	SDIO_D1_M0	GPIO2_A0_d	109	X
VO_LCDC_D13	I2S0_SDIO	SDIO_D2_M0	GPIO2_A5_d	114	X
VO_LCDC_D12	I2S0_SDO0	SDIO_D3_M0	GPIO2_A4_d	113	X
VO_LCDC_D10	I2S0_MCLK	SDIO_CLK_M0	GPIO2_A2_d	111	X
VO_LCDC_D11	I2S0_SDO3_SDI1	SDIO_CMD_M0	GPIO2_A3_d	112	X
VO_LCDC_D14	I2S0_SDO2_SDI2	UART0_RTS_M1	GPIO2_A6_d	107	X
VO_LCDC_D15	I2S0_SDO1_SDI3	UART0_CTS_M1	GPIO2_A7_d	106	X
VO_LCDC_D16	I2C1_SCL_M1	UART0_RX_M1	GPIO2_B0_d	105	X
VO_LCDC_D17	I2C1_SDA_M1	UART0_TX_M1	GPIO2_B1_d	104	X

Figure 2-26 RV1106G I2S0 module

The design recommendations for the pull-down and matching design of the I2S0 interface as show in the Tab.2-13.

Table 2-13 RV1103G/RV1106G I2S0 interface design

Signal	Internal pull-up and pull-down	Connection mode	Description (chipset)
I2S0_LRCK	pull down	Direct connect	I2S0 frame clock, for soundtrack choose
I2S0_SCLK	pull down	22ohm resistor in series	I2S0 bit clock
I2S0_SDIO	pull down	Direct connect	I2S0 data input channel0
I2S0_SDO0	pull down	Direct connect	I2S0 data output channel0
I2S0_MCLK	pull down	22ohm resistor in series	I2S0 system clock output
I2S0_SDO3_SDI1	pull down	Direct connect	I2S0 data output channel3 I2S0 data input channel1
I2S0_SDO2_SDI2	pull down	Direct connect	I2S0 data output channel2 I2S0 data input channel2
I2S0_SDO1_SDI3	pull down	Direct connect	I2S0 data output channel1 I2S0 data input channel3

2.3.6 VO Interface

The RV1106G chip has a built-in VOP, which supports RGB/MCU/BT1120/BT656 and other video output modes. The software needs to configure the corresponding output mode, and the hardware should also pay attention to the connection sequence of the signals.



Figure 2-27 RV1106G video output interface



NOTE

RV1103G doesn't support display interface..

Mode	RGB parallel		RGB serial		BT1120	BT656	mcu
	18bit	16bit	2x12	3x8/4x8	16bit	8bit	8bit
LCD_D0	B0	B0	D0	D0	D0	D0	D0
LCD_D1	B1	B1	D1	D1	D1	D1	D1
LCD_D2	B2	B2	D2	D2	D2	D2	D2
LCD_D3	B3	B3	D3	D3	D3	D3	D3
LCD_D4	B4	B4	D4	D4	D4	D4	D4
LCD_D5	B5	G0	D5	D5	D5	D5	D5
LCD_D6	G0	G1	D6	D6	D6	D6	D6
LCD_D7	G1	G2	D7	D7	D7	D7	D7
LCD_D8	G2	G3	D8	--	D8	--	--
LCD_D9	G3	G4	D9	--	D9	--	--
LCD_D10	G4	G5	D10	--	D10	--	--
LCD_D11	G5	R0	D11	--	D11	--	--
LCD_D12	R0	R1	--	--	D12	--	--
LCD_D13	R1	R2	--	--	D13	--	--
LCD_D14	R2	R3	--	--	D14	--	--
LCD_D15	R3	R4	--	--	D15	--	--
LCD_D16	R4	--	--	--	--	--	--
LCD_D17	R5	--	--	--	--	--	--
LCD_DEN	DEN	DEN	DEN	DEN	--	--	RDN
LCD_HSYNC	HSYNC	HSYNC	HSYNC	HSYNC	--	--	WRN
LCD_VSYNC	VSYNC	VSYNC	VSYNC	VSYNC	--	--	CSN
LCD_DCLK	DCLK	DCLK	DCLK	DCLK	DCLK	DCLK	RS

Support BT601 YCbCr 422 8bit output
 Support BT656 YCbCr 422 8bit output
 Support BT1120 YCbCr 422 16bit output, single-edge sampling
 Support RGB parallel 18bit/16bit output
 Support RGB serial 2x12/3x8/4x8 mode output
 Support Mcu serial 3x8/4x8 mode output

BT1120 16bit Mode
 Default: D0-D7 <--> Y0-Y7 , D8-D15 <--> C0-C7
 Swap ON: D0-D7 <--> C0-C7 , D8-D15 <--> Y0-Y7

Figure 2-28 RV1103G/RV1106G video output connect sequence

to avoid the damage caused by the surge.

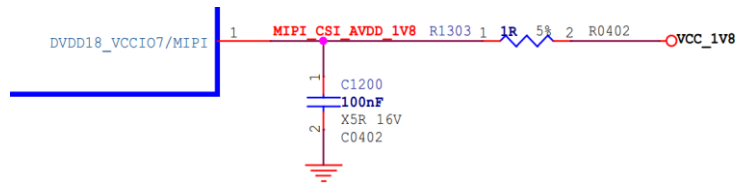


Figure 2-31 RV1103G/RV1106G MIPI CSI power supply series magnetic beads

2.3.7.2 VICAP Interface

RV1106G has 2 VI-CIF interfaces, VCCIO6 and VCCIO7. VI_CIF_M1 of VCCIO6 only supports 8/10bit parallel port and BT656 input, while VI_CIF_M0 of VCCIO7 supports 8/10/12 parallel port input, BT656 and BT1120 input. In the product design, the interface only supports the 1.8V working level, so the IO level/I2C pull-up level of the peripheral must be consistent with it, otherwise it will cause abnormal work or fail to work.

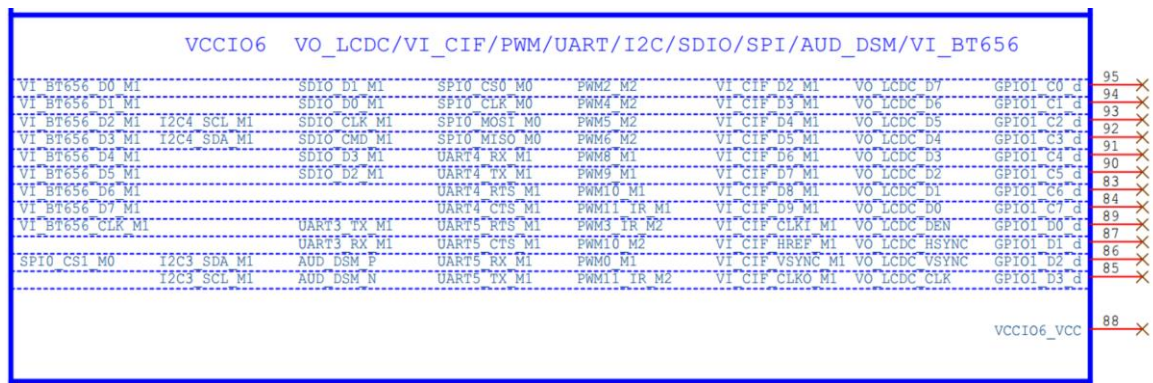


Figure 2-32 RV1106G VI-CIF module

Mode	VICAP_M0					VICAP_M1		
	CIF Parallel			BT1120	BT656	CIF Parallel		BT656
	8bit	10bit	12bit	16bit	8bit	8bit	10bit	8bit
VICAP_D0	--	--	--	D0	--	--	D0	--
VICAP_D1	--	--	--	D1	--	--	D1	--
VICAP_D2	--	--	--	D2	--	D0	D2	D0
VICAP_D3	--	--	--	D3	--	D1	D3	D1
VICAP_D4	--	--	D0	D4	--	D2	D4	D2
VICAP_D5	--	--	D1	D5	--	D3	D5	D3
VICAP_D6	--	D0	D2	D6	--	D4	D6	D4
VICAP_D7	--	D1	D3	D7	--	D5	D7	D5
VICAP_D8	D0	D2	D4	D8	D0	D6	D8	D6
VICAP_D9	D1	D3	D5	D9	D1	D7	D9	D7
VICAP_D10	D2	D4	D6	D10	D2	--	--	--
VICAP_D11	D3	D5	D7	D11	D3	--	--	--
VICAP_D12	D4	D6	D8	D12	D4	--	--	--
VICAP_D13	D5	D7	D9	D13	D5	--	--	--
VICAP_D14	D6	D8	D10	D14	D6	--	--	--
VICAP_D15	D7	D9	D11	D15	D7	--	--	--
VICAP_CLKIN	CLKIN	CLKIN	CLKIN	CLKIN	CLKIN	CLKIN	CLKIN	CLKIN
VICAP_HSYNC	HSYNC	HSYNC	HSYNC	--	--	HSYNC	HSYNC	--
VICAP_CLKOUT	CLKOUT	CLKOUT	CLKOUT	--	--	CLKOUT	CLKOUT	--
VICAP_VSYNC	VSYNC	VSYNC	VSYNC	--	--	VSYNC	VSYNC	--

Support BT601 YCbCr 422 8bit input
Support BT656 YCbCr 422 8bit input
Support RAW 8/10/12bit input
Support BT1120 YCbCr 422 16bit input, single/dual-edge sampling
Support 2/4 mixed BT656/BT1120 YCbCr 422 input

BT1120 16bit Mode
Default: D0-D7 <--> Y0-Y7 , D8-D15 <--> C0-C7
Swap ON: D0-D7 <--> C0-C7 , D8-D15 <--> Y0-Y7

Figure 2-33 RV1106G VI-CIF video input connect sequence

**NOTE**

RV1103G doesn't support VI CIF interface..

2.3.8 SARADC

The sampling range of SARADC is 0-1.8V with the accuracy is 10bits. The key array adopts a parallel type. The input key value can be adjusted by adding or subtracting keys and adjusting the ratio of the voltage divider resistance to realize multi-key input to meet customer product requirements. In the design, it is suggested that the value of any two buttons must be greater than +/-35, that is, the center voltage difference must be greater than 123mV.

The ADC_IN0 of the RV1103G/RV1106G chip is multiplexed as a RECOVER mode trigger (no need to update the LOADER), as shown in the figure below. On the premise that the system has already programmed the firmware, pull down ADCIN0 when the system starts, and keep SARADC_IN0 at 0V level, then RV1103G/RV1106G enters the Rockusb programming mode. When the PC recognizes the USB device, release the button to make ADCIN0 return to high level (1.8V), and then the firmware can be programmed.

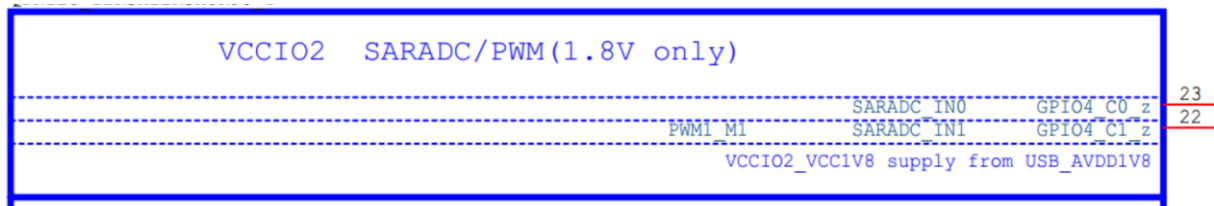


Figure 2-34 RV1103G/RV1106G SAR-ADC module

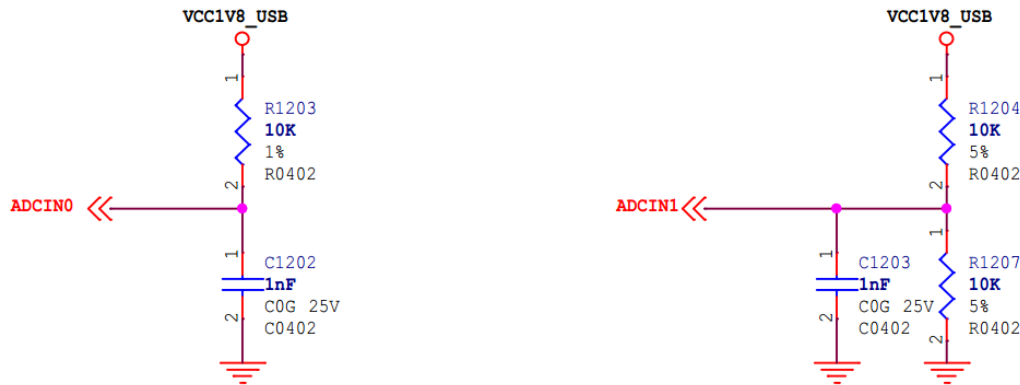


Figure 2-35 RV1103G/RV1106G SAR-ADC detection

2.3.9 UART Debug Circuit

The UATT2_M1 interface of RV1103G/RV1106G can be used as a debugging serial port. The UART2_M0 interface is multiplexed to the SDMMC interface, so if the product is debugged, the TF to USB adapter board can be used for debugging to avoid the need to disassemble the machine during debugging.

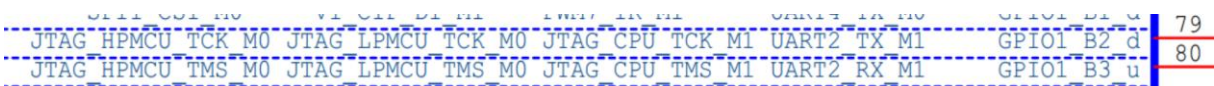


Figure 2-36 RV1103G/RV1106G UART2 multiplexing relationship M1

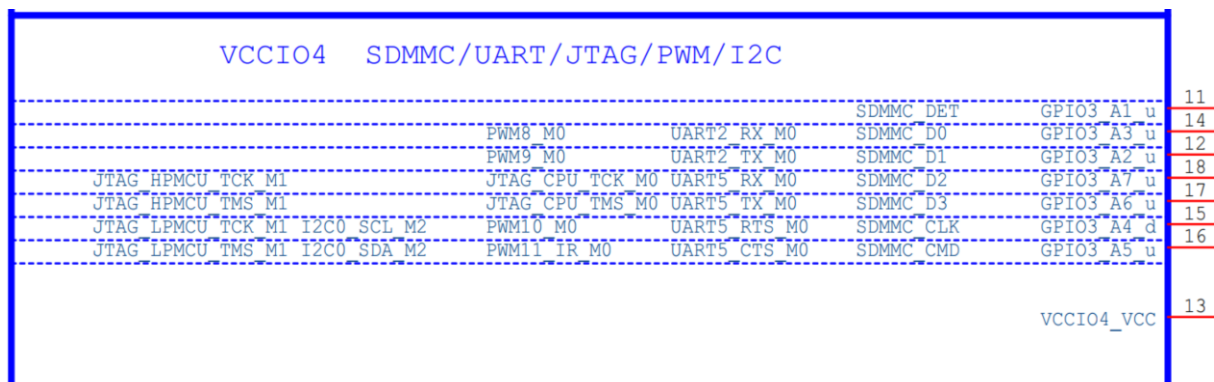


Figure 2-37 RV1103G/RV1106G UART2 multiplexing relationship M2

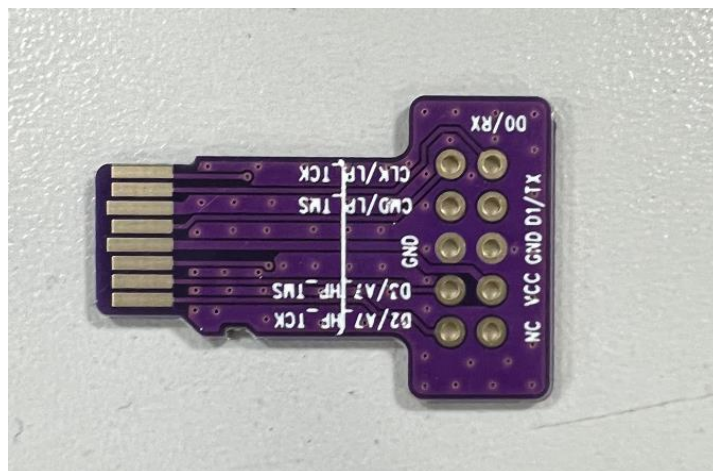


Figure 2-38 RV1103G/RV1106G TF to serial port conversion board

For the UART port number, please select the port number of the PC connected to the development board, the baud rate is 115200, and the flow control RTS/CTS does not need to be checked. If the built-in DB-9 port on the PC does not support high-speed mode, please use the USB-to-serial method.

波特率(B) :	115200
数据位(D) :	8
校验位(P) :	无
停止位(S) :	1
流控制(F) :	无

Figure 2-39 RV1103G/RV1106G serial port configuration

3 PCB Design Recommendations

3.1 PCB Stackup Design

In order to reduce the reflection phenomenon in the process of high-speed signal transmission, impedance matching must be maintained on the signal source, the receiving end and the transmission line. The impedance of a single-ended signal line depends on its line width and its relative position to the reference layer. The line width/line spacing between the differential pairs required for a specific impedance depends on the selected PCB stackup. Since the minimum line width and minimum line distance depend on the PCB type and cost requirements, due to this limitation, the selected PCB stackup must be able to achieve all impedance requirements on the board, including inner and outer layers, single-ended and differential lines, etc.

In RV1103G/RV1106G design, it's recommended to use 4-layer PCB stackup. Please provide the corresponding impedance requirements to the PCB manufacturer and design as required.

3.2 General Design Recommendation

(1) Trace length should include package and via.

(2) Intra-pair skew is the term used to define the difference length between + and - in a differential pair. Inter-pair skew is used to define the difference length between a differential pair and another differential pair. The space between signals is defined as air-gap distance.

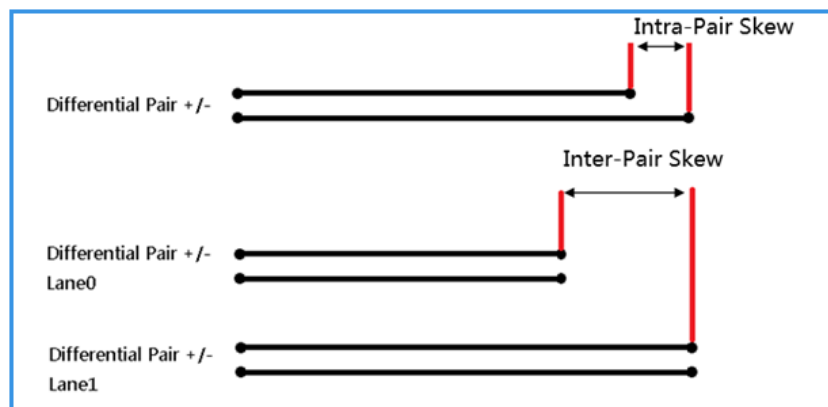


Figure 3-1 Differential Pair Skew

(3) Route trace over continuous GND plane without interruption. Any discontinuity or split on the reference plane will degrade signal integrity significantly.

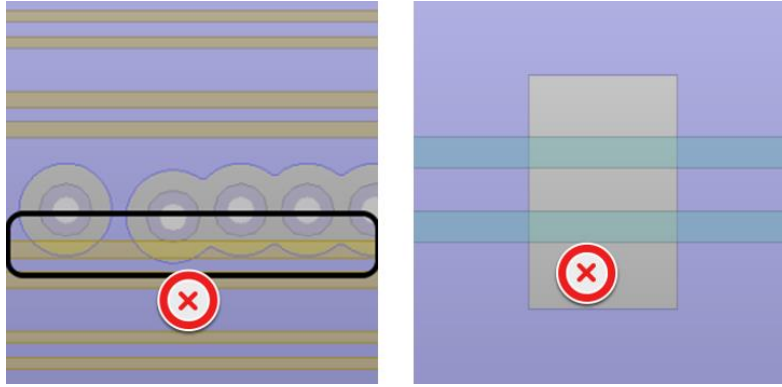


Figure 3-2 Discontinuous Reference Plane

(4) Voiding the ground plane underneath the SMT signal pad (as shown in the following figure red rectangle area) is necessary to minimize the impedance mismatch. SMT components include AC capacitor, ESD, CMC (common mode choke), connector, etc. If no specific requirement is provided, keep the voiding size as same as the pad or slightly bigger.

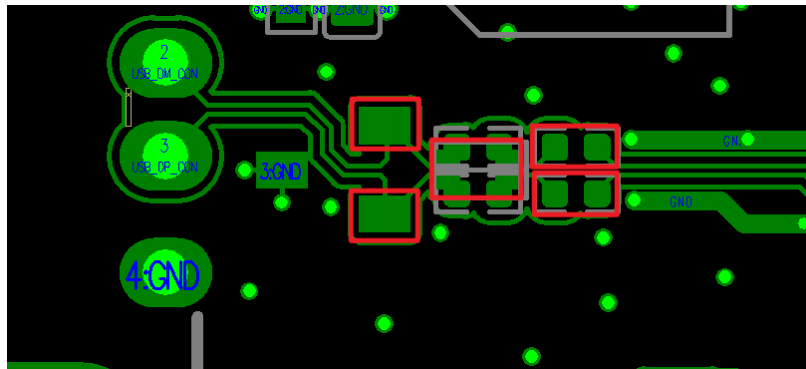


Figure 3-3 Reference Plane Voiding

(5) Recommend keeping traces ≥ 3 times the width of trace from GND plane on same layer.

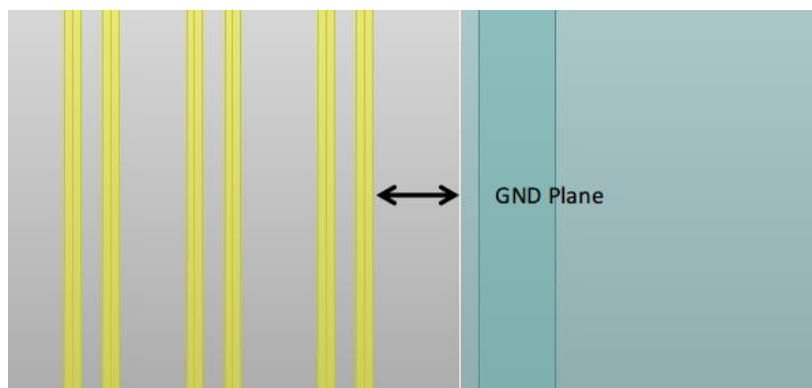


Figure 3-4 Airgap Between Trace and Plane

(6) Avoid routing across different reference planes. Recommend that high-speed differential signals are routed ≥ 40 mils from the edge on the reference plane.

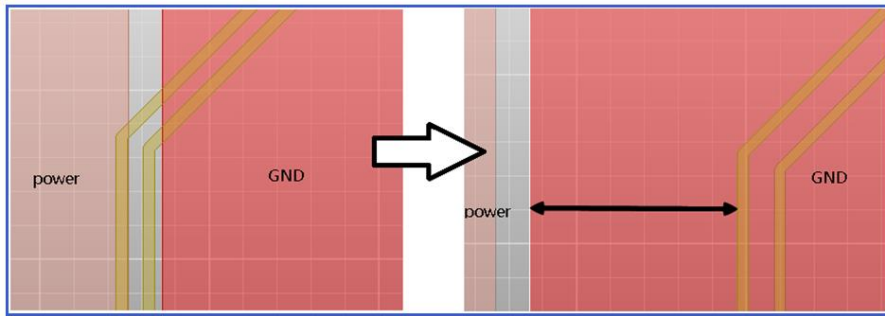


Figure 3-5 Reference Plane Edge

- (7) It's not recommend to place test point on any high-speed signals.
- (8) Bend should be minimized. If bends are needed, use 135° bends instead of 90°.
- (9) Serial resistor recommend to be placed near transmitter.
- (10) Each GND PAD at IC(such as emmc device、flash) recommend place a GND PTH Via.

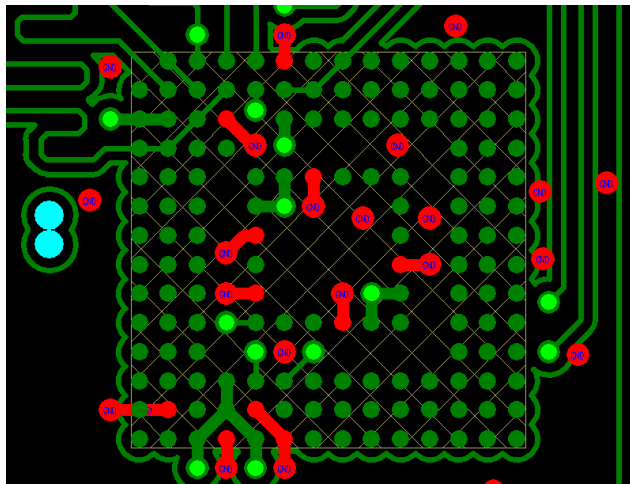


Figure 3-6 GND Via Placement

- (11) Avoid routing under or near crystals, oscillators, clock signal generators, switching regulators, mounting holes, magnetic devices or IC's that used for duplicate clock signals.
- (12) Remove all non-functional via pads.
- (13) Recommend each ESD GND pad has a GND PTH via, and keep the length from pad to via as short as possible.

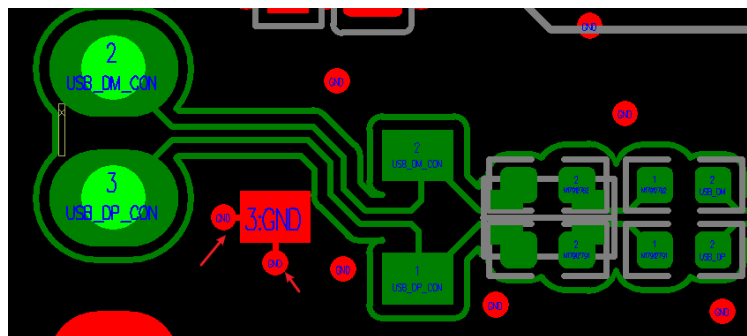


Figure 3-7 GND Via Placement for ESD Component

(14) Differential signals require equal lengths within pairs, that is, the time delay difference between P and N should be as small as possible. Therefore, when the time delay difference occurs between the differential lines P and N, the nearby winding is compensated. Special attention should be paid to the size of the winding, which should meet the requirements shown in the figure below to reduce the impact of sudden changes in impedance.

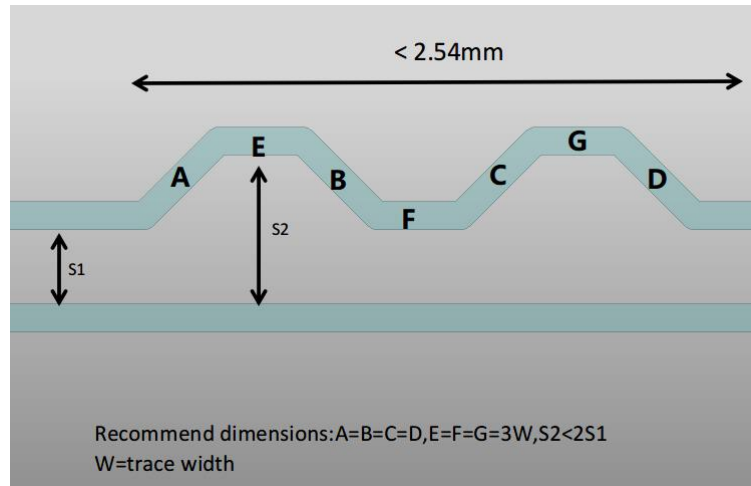


Figure 3-8 Winding Dimension

(15) If there is an unequal length (within 300mil) in the differential pair, make the compensation as soon as possible.

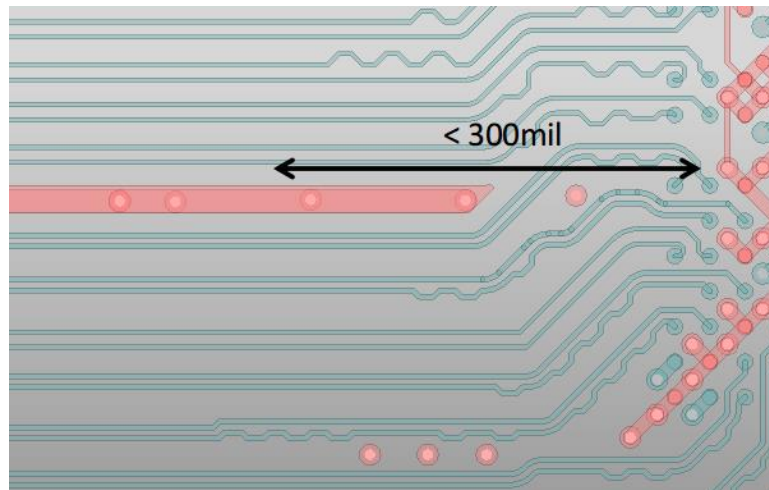


Figure 3-9 Skew Compensation

(16) Place ground stitching vias near the signal transition vias. For differential signals, both signal vias and stitching vias should be placed symmetrically.

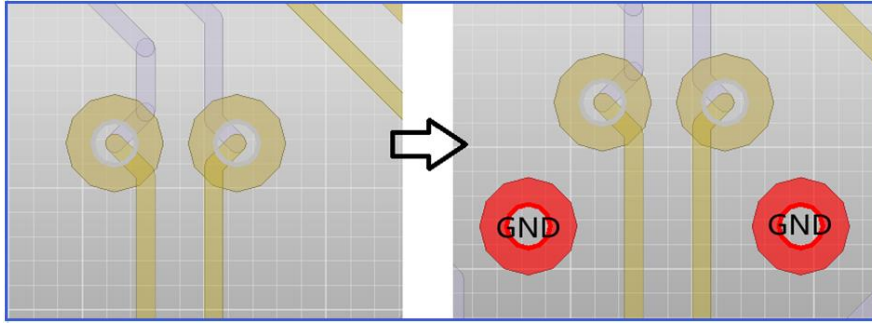


Figure 3-10 Gnd Via for Differential Via

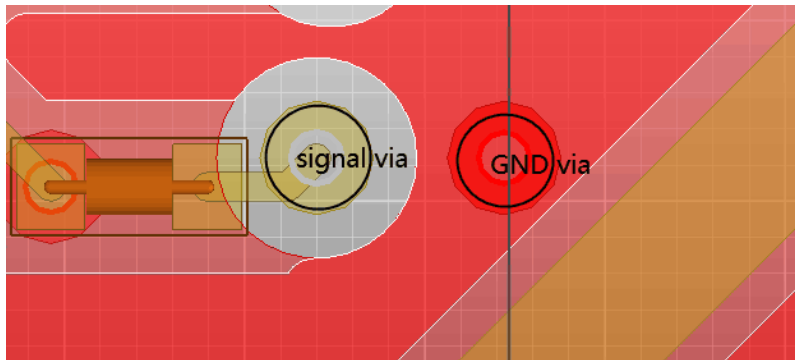


Figure 3-11 Gnd Via for Single-ended Via

(17) Route differential pairs on symmetry.

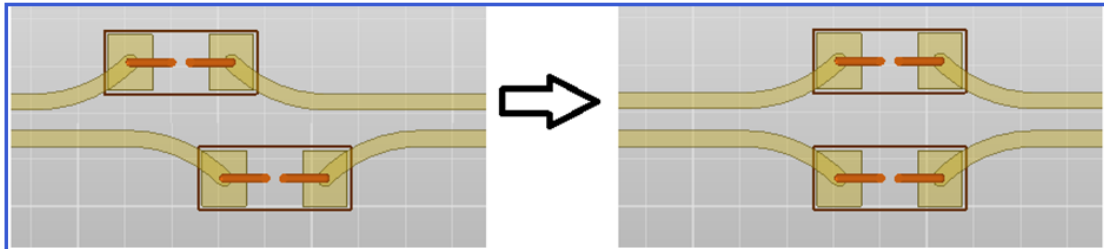


Figure 3-12 symmetry layout

(18) It is recommended to place at least one ground through via in each ground pad of the high-speed connector, and the through via should be as close as possible to the pad.

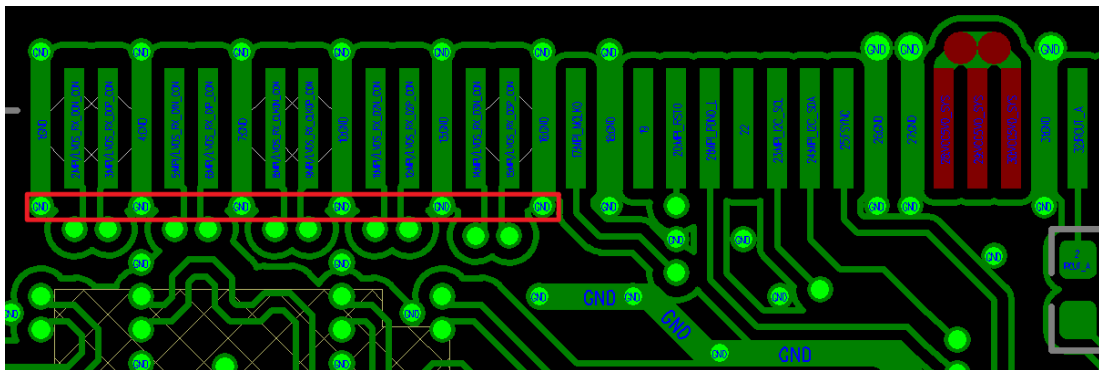


Figure 3-13 Gnd Via Placement

(19) Ground shape at connector near GND PAD recommend $\leq$ PAD size.

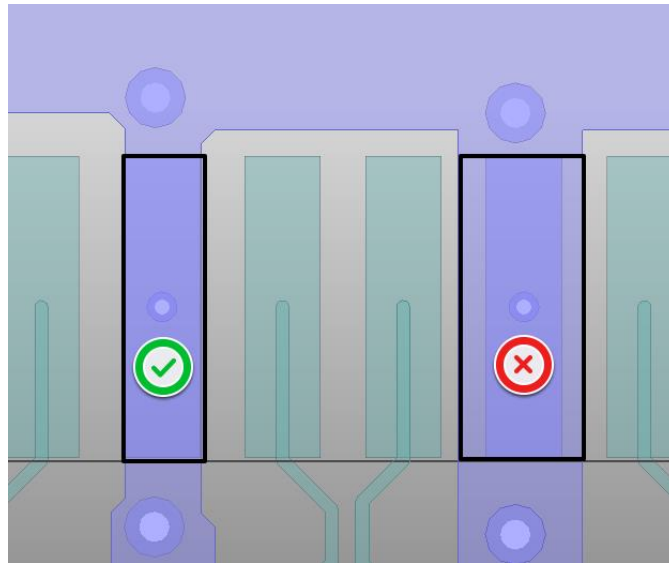


Figure 3-14 Ground Shape Near Connector Pad

(20) Recommend distance between ground plane and pad ≥ 3 times the width of trace at connector.

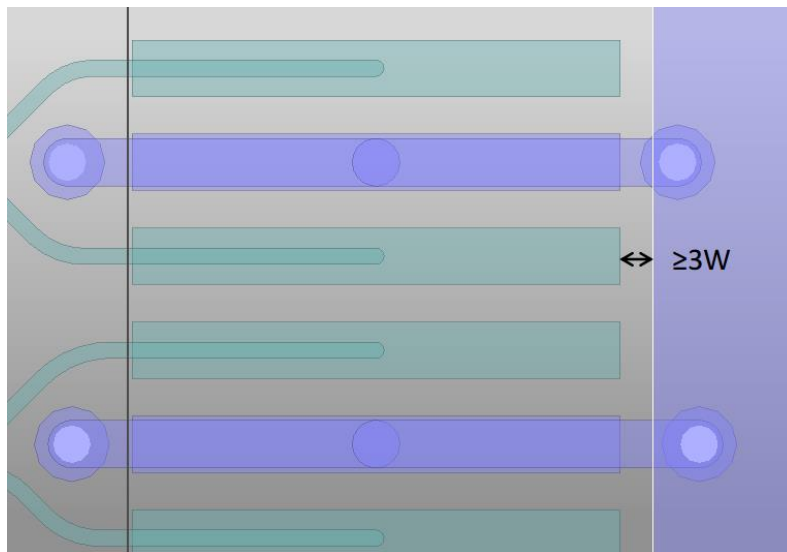


Figure 3-15 Airgap Between Pad to Plane

(21) Proposed ground shielding.

L: GND via interval length.

D: Distance from shielding to trace $\geq 3*W$.

(24) Minimize the stub length as much as possible, it is recommended to keep the stub length as zero.

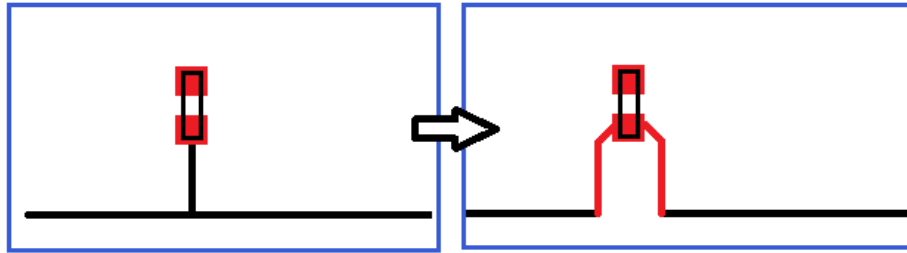


Figure 3-19 Trace Stub

(25) the signal can't be routed over two different reference planes.

3.3 Interfaces Design Guide

3.3.1 SDMMC/SDIO PCB Design

SDMMC0/SDIO signal design requirements as follow:

- The length of SDMMC0_DATA/SDIO_DATA, SDMMC0_CMD/ SDIO_CMD is based on the length of SDMMC0_CLK/SDIO_CLK, and the bias $\leq 120\text{mil}$.
- The signal takes GND as the reference plane.
- SDMMC0 connect TF card. For ESD devices, select devices with a junction capacitance of less than 1pF, and it is recommended to place them close to the connector.
- The length of package + PCB line should not exceed 4 inches.
- It is recommended that the SDMMC_DATA/CMD traces be grounded in the whole group, and the SDMMC_CLK is recommended to be grounded separately. See (24) of section 3.2 “General Design Recommendations” for details.



Figure 3-20 RV1103G/RV1106G SDMMC layout recommendation

Table 3-1 SDMMC/SDIO Layout Requirements

Parameter	Requirement
Trace Impedance	50Ω ±10% single ended
Data to clock matching	<120mil
Max trace length	<4 inches
The minimum spacing of SDMMC Signals	At least 2 times the width of SDMMC trace.

3.3.2 FEPHY PCB Design

The FEPHY signal design requirements are as follows:

- The Intra-pair skew is controlled within ±30mil, and the differential impedance is controlled within 100ohm ±10%.
- The differential signal must use GND as the reference plane. Maintain the integrity of the signal reference plane.
- Avoid adjacent to other signals, and ensure that the distance from other signals is greater than 20mil.
- It is recommended to use TVS tube as the protection device, the breakdown voltage is 8kV, and the response time is less than 1ns.

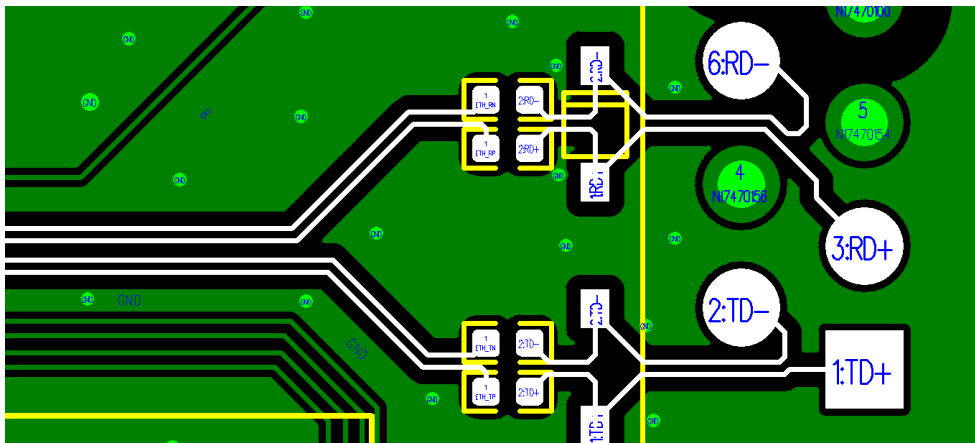


Figure 3-21 RV1103G/RV1106G FEPHY layout recommendations

Table 3-2 FEPHY Layout Requirements

Parameter	Requirement
Trace Impedance	100Ω ±10% differential
Max intra-pair skew	<30mil
Max trace length on carrier board	<6 inches
Maximum allowed via	Recommend less than 4 vias Cannot exceed 6 vias

3.3.3 USB PCB Design

The USB signal design requirements are as follows:

- The Intra-pair skew is controlled within $\pm 30\text{mil}$, and the differential impedance is controlled within $90\Omega \pm 10\%$.
- The differential signal must use GND as the reference plane. Maintain the integrity of the signal reference plane.
- The trace length of the USB2.0 differential signal line of the RV11XX should not exceed 6 inches, and the number of vias should not exceed 6. A GND via needs to be placed near the USB signal via for better signal quality.
- Avoid adjacent to other signals, and ensure that the distance from other signals is greater than 20mil.
- The parasitic capacitance of the ESD device is recommended to be less than 1pF.

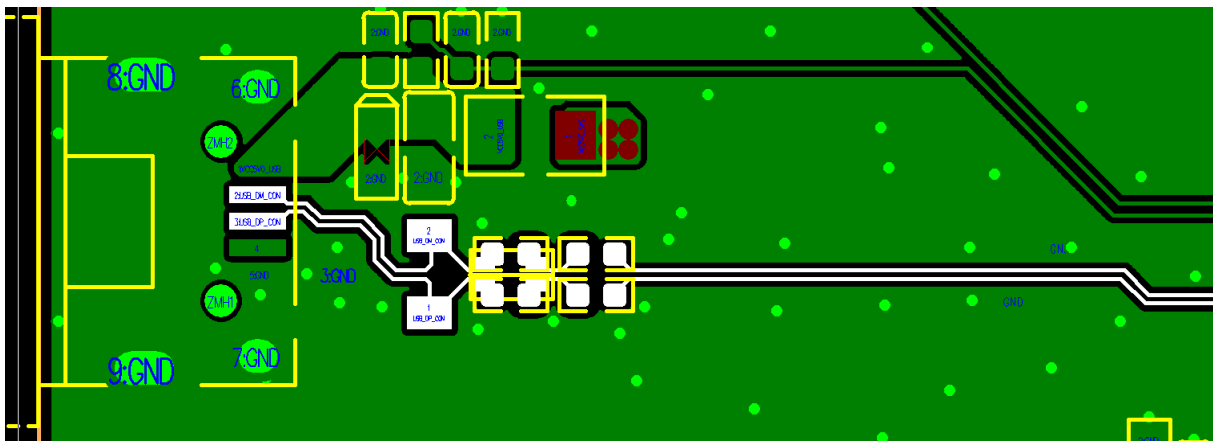


Figure 3-22 RV1103G/RV1106G USB layout recommendations

Table 3-3 USB Layout Requirements

Parameter	Requirement
Trace Impedance	$90\Omega \pm 10\%$ differential
Max intra-pair skew	$< 30\text{mil}$
Max trace length on carrier board	< 6 inches
Maximum allowed via	Recommend less than 4 vias Cannot exceed 6 vias

3.3.4 Audio PCB Design

The ACODEC signal design requirements are as follows:

- The MIC differential signal traces should be grounded and kept away from other high-speed signals.
- The LINEOUT signal needs to be grounded separately and away from other high-speed signals.
- The VCM capacitor is placed close to the chip.

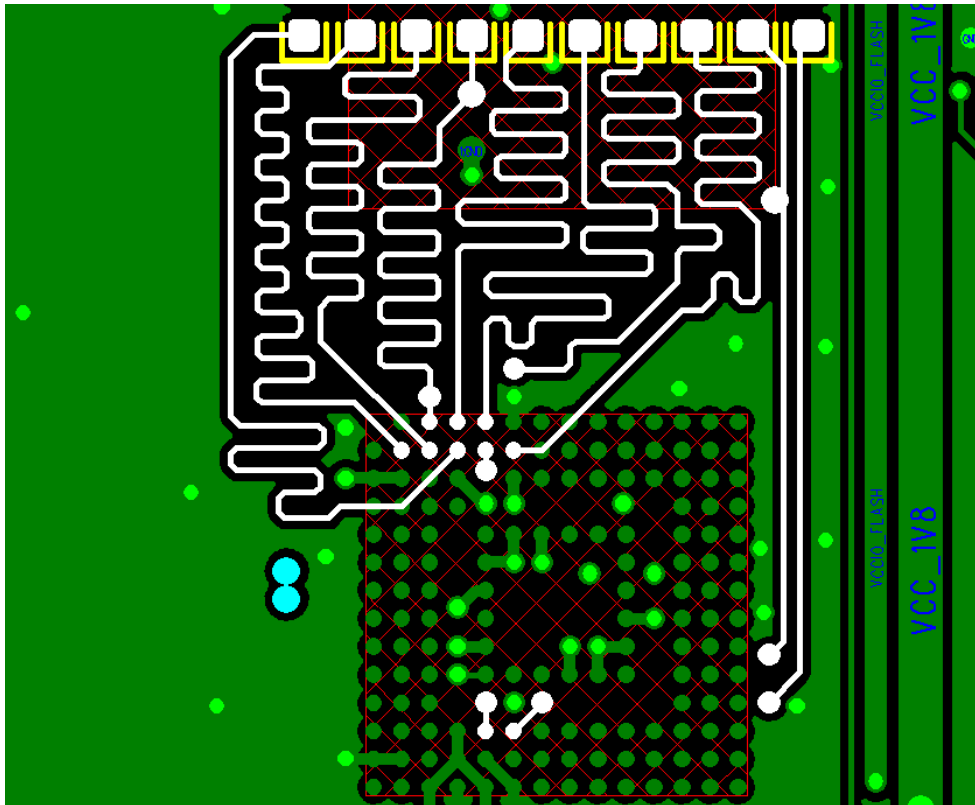


Figure 3-24 RV1106G EMMC layout recommendations

Table 3-5 EMMC Layout Requirements

Parameter	Requirement
Trace Impedance	50Ω ±10% single ended
Data to clock matching	<120mil
Max trace length	<4 inches
The minimum spacing of EMMC Signals	At least 2 times the width of EMMC trace.
The minimum spacing between EMMC and other Signals	Recommend 3 times the width of EMMC trace. At least 2 times the width of EMMC trace.

3.3.6 Video Input PCB Design

MIPI CSI/LVDS RX signal design requirements:

- The differential signal takes GND as the reference plane and keeps the reference plane intact;
- The PCB trace length is recommended to be within 7.2 inches, the P/N equal length of the differential pair is controlled within 24 mil, and the sampling differential clock is used as the reference between the pairs, and the equal length is controlled within 42 mil (package + PCB joint control equal length).
- The PCB traces of the MIPI RX differential pair control the differential impedance of $100\Omega \pm 10\%$.
- When differential signals pass through the connector, GND pins must be used for isolation between

adjacent differential signal pairs.

- To improve MIPI-CSI performance, please place the decoupling capacitors of the controller power supply close to the pins;
- The distance between the differential pair and other signals follows the 3W principle.

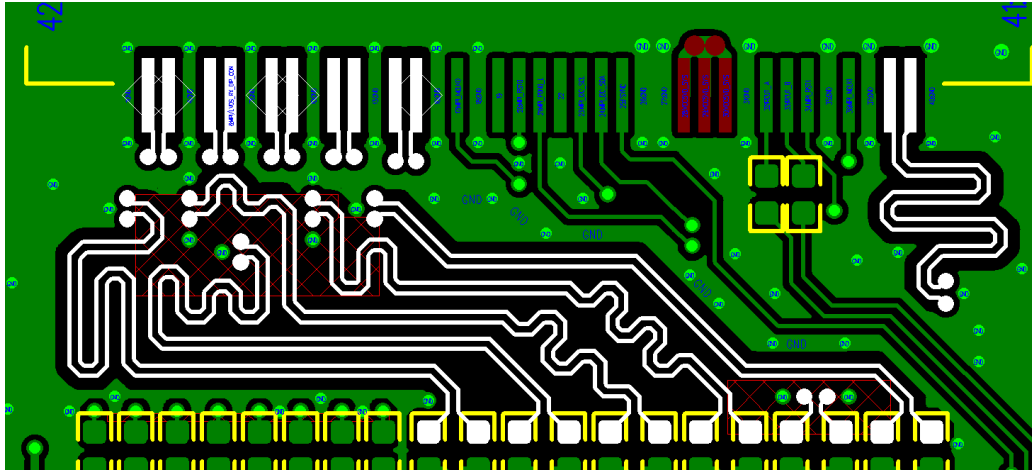


Figure 3-25 RV1103G/RV1106G MIPI CSI/LVDS RX layout recommendations

Table 3-6 MIPI CSI/LVDS RX Layout Requirements

Parameter	Requirement
Trace Impedance	100Ω ±10% differential
Data to clock matching	<120mil
Max trace length	<7.2 inches
The minimum spacing of EMMC Signals	At least 2 times the width of EMMC trace.
The minimum spacing between EMMC and other Signals	Recommend 3 times the width of EMMC trace. At least 2 times the width of EMMC trace.

Parallel CIF interface signal design requirements:

- Avoid signal traces crossing the power split area, and keep the signal reference plane intact;
- The spacing between adjacent signal traces should be kept to the principle of "3W";
- The line lengths of CIF_DATA, CIF_HSYNC, and CIF_VSYNC are based on the line lengths of CIF_CLKOUT and CIF_CLKIN, and the deviation is controlled within ±50mil.

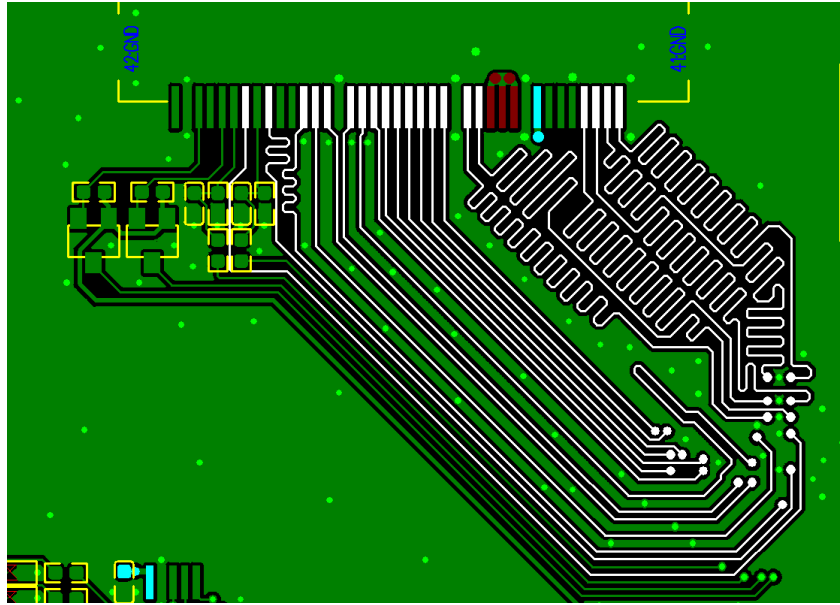


Figure 3-26 RV1106G Parallel CIF layout recommendations

Table 3-7 Parallel CIF Layout Requirements

Parameter	Requirement
Trace Impedance	100Ω ±10% differential
Data to clock matching	<120mil
Max trace length	<7.2 inches
The minimum spacing of EMMC Signals	At least 2 times the width of EMMC trace.
The minimum spacing between EMMC and other Signals	Recommend 3 times the width of EMMC trace. At least 2 times the width of EMMC trace.

3.3.7 Video Output PCB design

Video output BT1120 and LCDC signal design requirements:

Table 3-8 BT1120 Layout Requirements (≤74.25Mhz)

Parameter	Requirement
Trace Impedance	50Ω ±10% single ended
Data to clock matching	<180mil
Max trace length	<5 inches
The minimum spacing of LCDC Signals	Recommend 2 times the width of LCDC trace.

Table 3-9 LCDC Layout Requirements (≤74.25Mhz)

Parameter	Requirement
Trace Impedance	50Ω ±10% single ended
Data to clock matching	<180mil
Max trace length	<5 inches
The minimum spacing of LCDC Signals	Recommend 2 times the width of LCDC trace.

4 Thermal Design Suggestion

Unfinished

5 ESD/EMI Protection Design

5.1 Overview

This chapter provides ESD/EMI protection design suggestion for RV1103G/RV1106G product design to help customers to improve anti-static and anti-electromagnetic interference ability of product.

5.2 Terms Interpretation

Terms of this chapter are explained as below:

- ESD: Electro-Static discharge
- EMI: Electromagnetic Interference, including conduction interference and radiation interference

5.3 ESD Protection

- Ensure reasonable mold design; anti-ESD devices should be reserved for ports and connectors.
- Protect and isolate sensitive components in PCB layout.
- Try best to put the RK3588 and core components in the center of PCB layout. If it may not be able to put them in the center, ensure that the shielding cover has 2mm distance at least from the board edge and is connected to GND safely
- PCB layout is based on function module and signal flow direction, sensitive components should be mutually independent, and it is better to isolate the parts that are easy to produce interference;
- Place ESD components reasonably. Generally, place them near the source, that is, place ESD components in the junction or where electrostatic discharge.
- The layout of the components should be far away from the edge of the board and a certain distance from the connector.
- The PCB surface should have a good GND loop, and each connector should be with good GND connection loop on the surface. If there is a shielding cover, it should be connected to the surface ground as much as possible, and drill as much ground vias at the soldering place of the shielding cover as possible. In order to do this, it is required that each connector should not be routed on the surface, and there should not be traces with surface copper cut off in a large area;
- Do not go through the edge of surface layer and make as many ground vias as possible.
- Isolate signal from ground if necessary.
- Expose GND copper of PCB as much as possible, in order to strengthen the electrostatic discharge effect, or to facilitate to add foam and other remedial measures

5.4 EMI Protection

- Electromagnetic interference has three factors: interference sources, coupling channels and sensitive devices. We have no way to deal with sensitive devices, so EMI problem can only start with interference sources and coupling channels. The best way to resolve EMI issues is to eliminate

interference source. If it cannot eliminate, try to cut off coupling channels or avoid antenna effect;

- It is difficult to eliminate interference source on PCB. We can take actions such as filtering, grounding, balancing, resistance controlling, improving signal quality (e.g. termination connection) etc. Generally, several methods will be applied together, but the basic requirement is good grounding;
- The commonly used EMI materials include shielding cover, special filter, resistor, capacitor, inductor, magnetic bead, common mode choke/magnetic ring, wave-absorbing material, spread frequency device, etc.;
- The rules to select filters: if the load (receiver) is high resistance (regular single ended signal interface is high resistance, such as SDIO, RGB, CIF etc.), select capacitive filter components and parallel connect to circuit; if the load (receiver) is low resistance (such as power output interface), select inductive filter component and serial connect to circuit. After using the filter device, the signal quality cannot exceed its SI permission. Differential interface usually uses common mode choke to suppress EMI;
- The shielding measures on PCB should have good grounding, otherwise it will cause radiation leakage or form antenna effect. The shielding of connectors should comply with relevant technical standards;
- EMI has the same high requirement as ESD on layout. The ESD Layout requirements described above are mostly suitable for EMI protection. Besides, add the following requirements:
 - Try best to ensure the integrity of the signal;
 - Differential line should in equal length and be tight coupling to ensure the symmetry of the differential signal, minimize the misplacement of differential signals to avoid EMI problems caused by phase mismatch;
 - Components with metal shell such as plug-in electrolytic capacitors should avoid coupling interference signals to radiate. Also need to avoid component interference signals coupling from shell to other signal lines.

6 Soldering Process

Unfinished